CollectData [SimTalk] - CostAnalyzer

Sets if the *CostAnalyzer* designated by <Path> collects data (`true`) or does not collect data (`false`).

Type

Attribute

Syntax

```
<Path>.CollectData:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MyCostAnalyzer.CollectData := true
```

See also

Collect Data [check box] - CostAnalyzer

HtmlReport

HtmlReport [object]

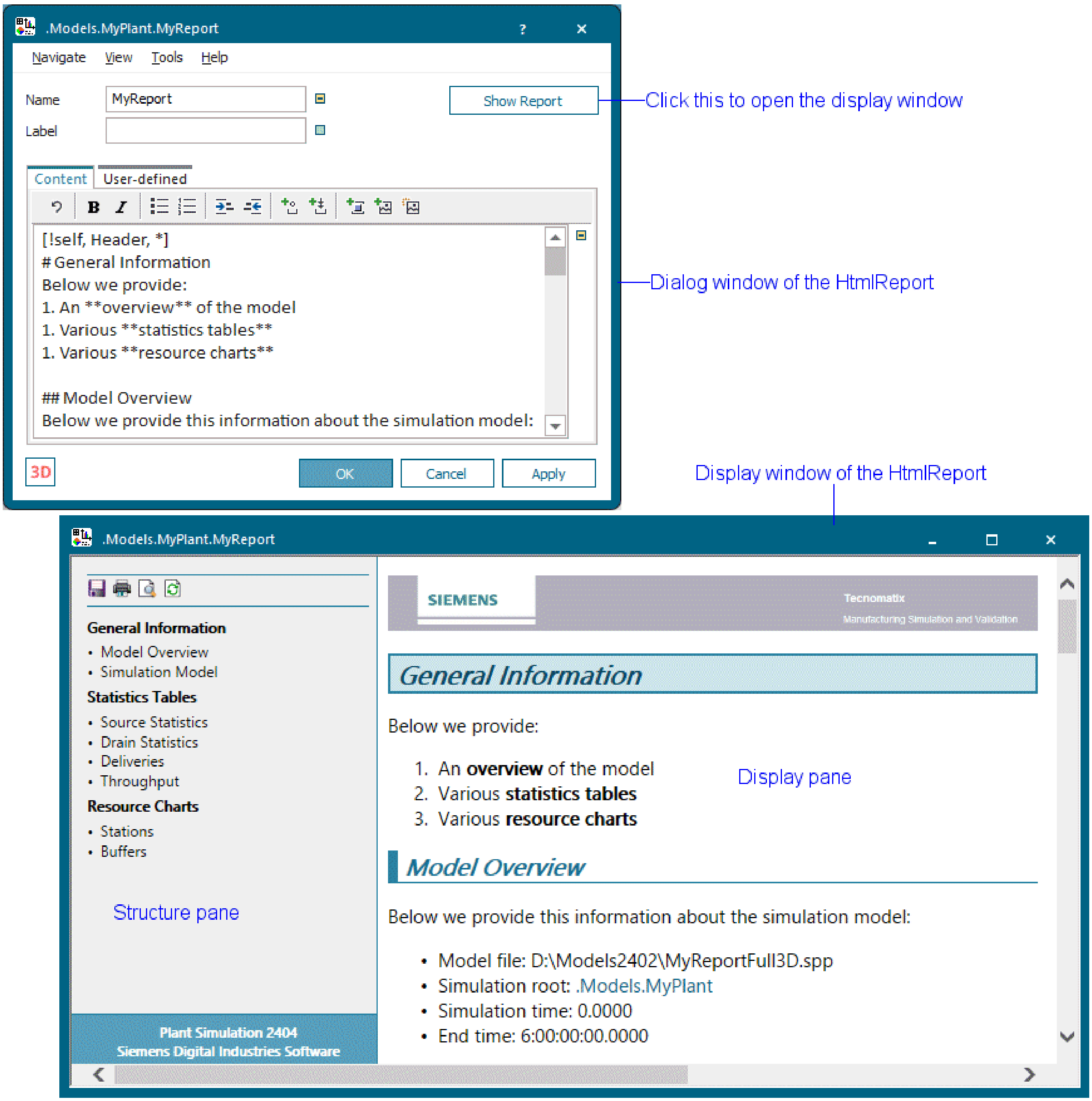
Use the object *HtmlReport* for presenting current data and results of the simulation run in a report that you can share with co-workers and customers.

Image

Description

You can save the *HtmlReport* as an .htm file and open this file in an HTML-Browser, such as *Microsoft Edge*, *Firefox*, *Google Chrome*, etc. *Plant Simulation* is not required to open the *HtmlReport* in an external Browser, meaning that the *HtmlReport* can be displayed on any computer on which an HTML-Browser is installed.

> **Note**
>
> The easiest way to configure the contents of the *HtmlReport* is to **drag** an object from the *Frame* to the tab **Content** and to **drop** it there. You can type in additional content and formatting settings in the built-in markup language on the **Tab Content**.



You can also show another *HtmlReport* within an *HtmlReport*, compare **Display a HtmlReport**.

To show a **tooltip** with information about the *HtmlReport*, hover with the mouse over it.

To change the length of the graphic and the anchor points of the *HtmlReport*, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model

To add the object *HtmlReport* to your simulation model, click **Manage Class Library** > **Basic Objects > UserInterface > HtmlReport** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.

See also

Show Statistics and Other Values in a Report

Show Reports in a Hierarchically Structured Model

Dialog Box of the HtmlReport

Dialog Box of the HtmlReport

Double-click the icon of the *HtmlReport* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under **Dialog Items of the Objects**.

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box **Edit 3D Properties**:

- Click the button **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.

See also

Show Report

Show Report

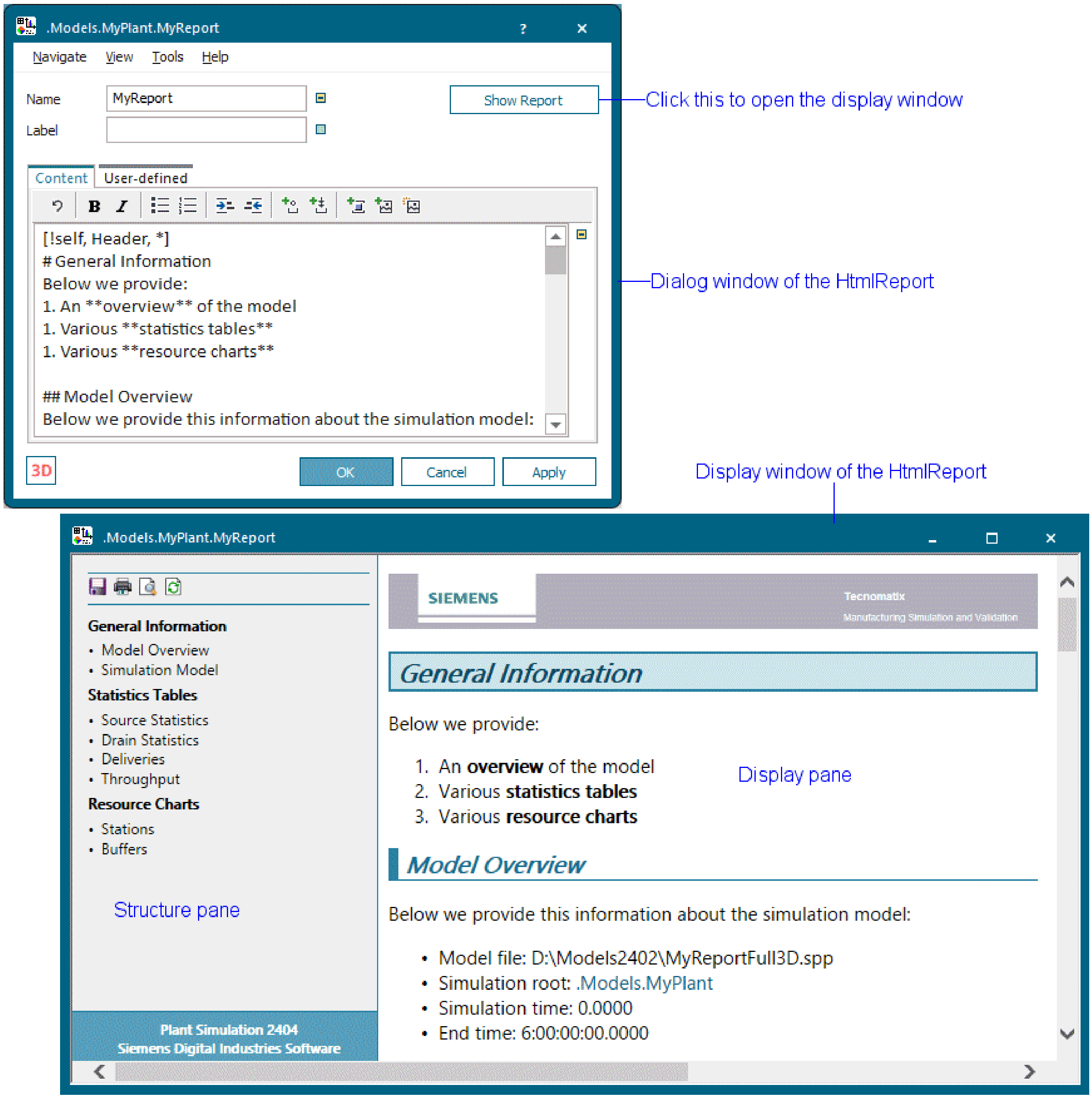
Show Report

To open the display window of the *HtmlReport*, click **Show Report**.

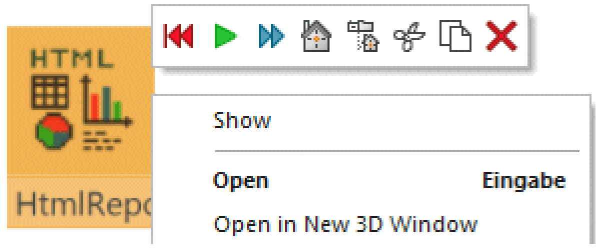
Remarks

The **Display Window** contains:

- The **Display Pane of the Display Window** on the right-hand side shows the content of the report which you define on the **Tab Content**.
- The **Structure Pane of the Display Window** on the left-hand side shows the first two heading levels of the report. Click a heading in the structure pane to jump to this location in the display pane.



Instead, you can also right-click the *HtmlReport* object in the *Frame* and select **Show** on the context menu.



SimTalk

IsShown [SimTalk] - HtmlReport

show [SimTalk]

Content [SimTalk] - HtmlReport

WindowHeight [SimTalk]

WindowWidth [SimTalk]

Structure Pane of the Display Window

The structure pane of the display window shows the table of contents of your *HtmlReport*.

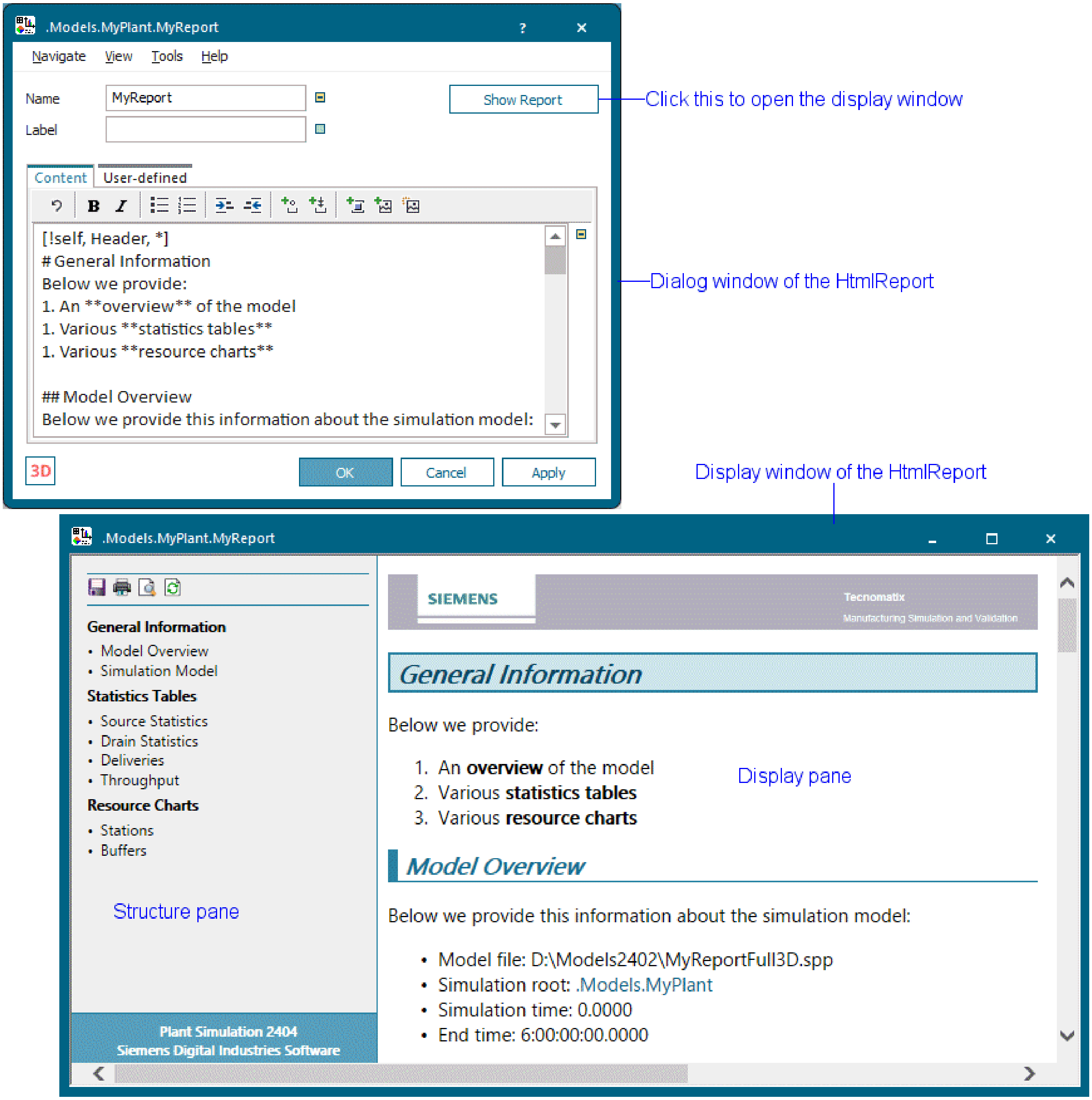
Remarks

Proceed as follows to do the following in the structure pane of the display window:

- To **jump** to a certain heading of level 1 or 2 in the display window, click this heading in the structure pane.
- To **print** the report, click **Open the print dialog**.

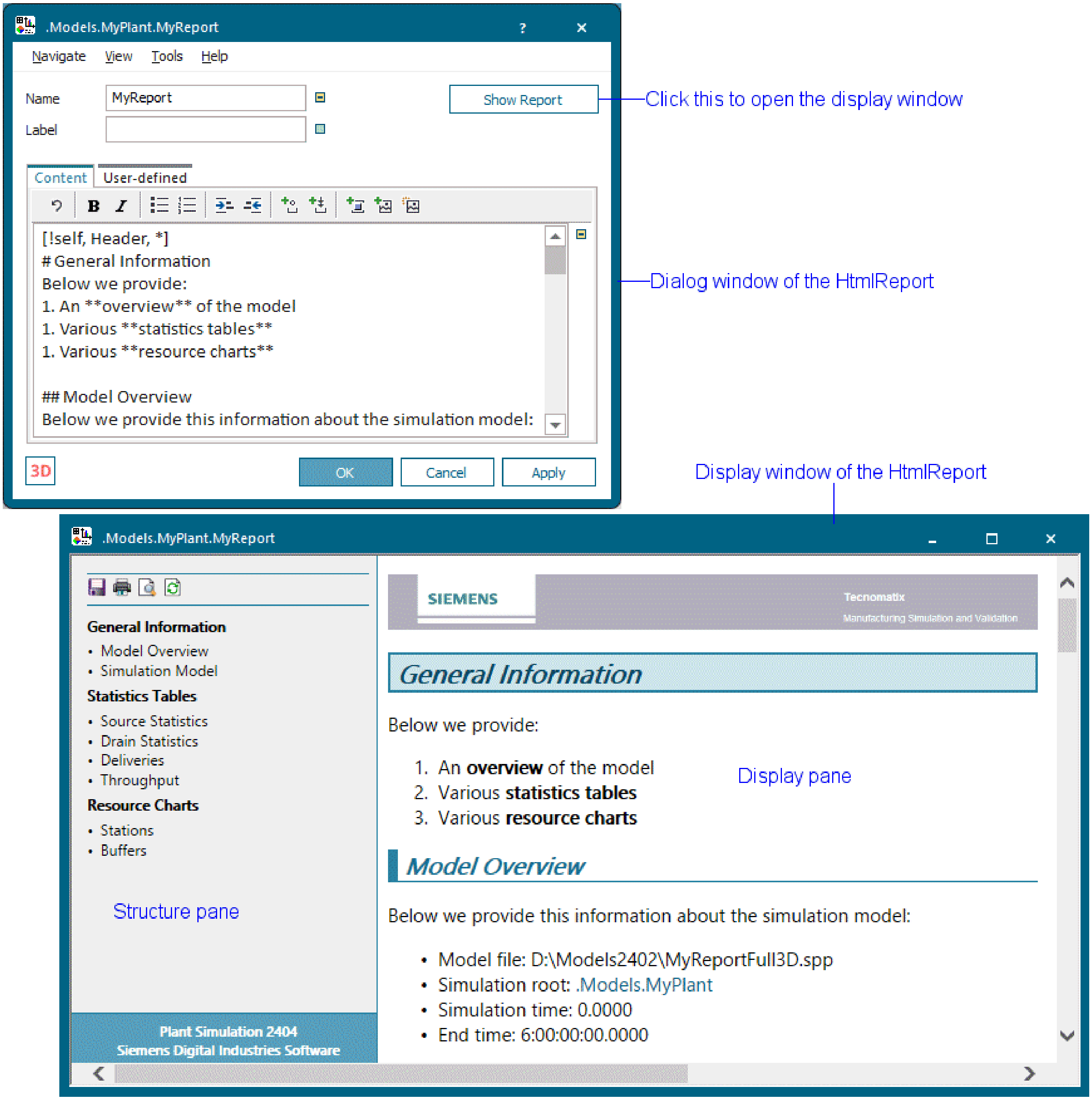
> **Note**
>
> **Print** and **Save** do not work if the local security policies deny access to the operating system within the HTML functionality.

- To **save** the content of the HTML page as a **Web Page, complete**, i.e., as an .htm file or an .html file, click **Save the report to an HTML file**.
- To **update** the displayed data with the current data of the model, click **Refresh the report**.
- To **scroll** the content of the structure pane and of the display pane **up**, roll the mouse wheel forward.
- To **scroll** the content of the structure pane and of the display pane **down**, and roll the mouse wheel backward.
- To **zoom** the content of the structure pane and of the display pane **in**, hold down **Ctrl** and roll the mouse wheel forward.
- To **zoom** the content of the structure pane and of the display pane **out**, hold down **Ctrl** and roll the mouse wheel backward.


.Models.MyPlant.MyReport
Navigate View Tools Help
Name MyReport
Label
Show Report
Click this to open the display window
Content User-defined
[!self, Header, *]
General Information
Below we provide:
1. An **overview** of the model
1. Various **statistics tables**
1. Various **resource charts**
Model Overview
Below we provide this information about the simulation model:
Dialog window of the HtmlReport
3D
OK Cancel Apply
Display window of the HtmlReport
.Models.MyPlant.MyReport
General Information
Model Overview
Simulation Model
Statistics Tables
Source Statistics
Drain Statistics
Deliveries
Throughput
Resource Charts
Stations
Buffers
Structure pane
Plant Simulation 2404
Siemens Digital Industries Software
SIEMENS
General Information
Below we provide:
1. An overview of the model
2. Various statistics tables
3. Various resource charts
Display pane
Model Overview
Below we provide this information about the simulation model:
Model file: D:\Models2402\MyReportFull3D.spp
Simulation root: .Models.MyPlant
Simulation time: 0.0000
End time: 6:00:00:00.0000

Display Pane of the Display Window

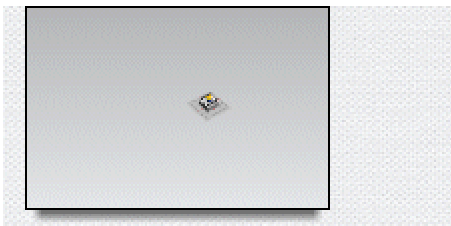
The **display pane** of the display window displays the content of the report which you defined on the tab **Content**.



Remarks

Proceed as follows to do the following in the structure pane of the display window:

- To **scroll** the content of the structure pane and of the display pane **up**, roll the mouse wheel forward.
- To **scroll** the content of the structure pane and of the display pane **down**, and roll the mouse wheel backward.
- To **zoom** the content of the structure pane and of the display pane **in**, hold down **Ctrl** and roll the mouse wheel forward.
- To **zoom** the content of the structure pane and of the display pane **out**, hold down **Ctrl** and roll the mouse wheel backward.
- To **open** the respective *Frame* or the respective *object* of a picture shown in the report, click the link of the caption line.



.Models.MyEnginePlant.SingleProc

SimTalk

save [SimTalk] - HtmlReport

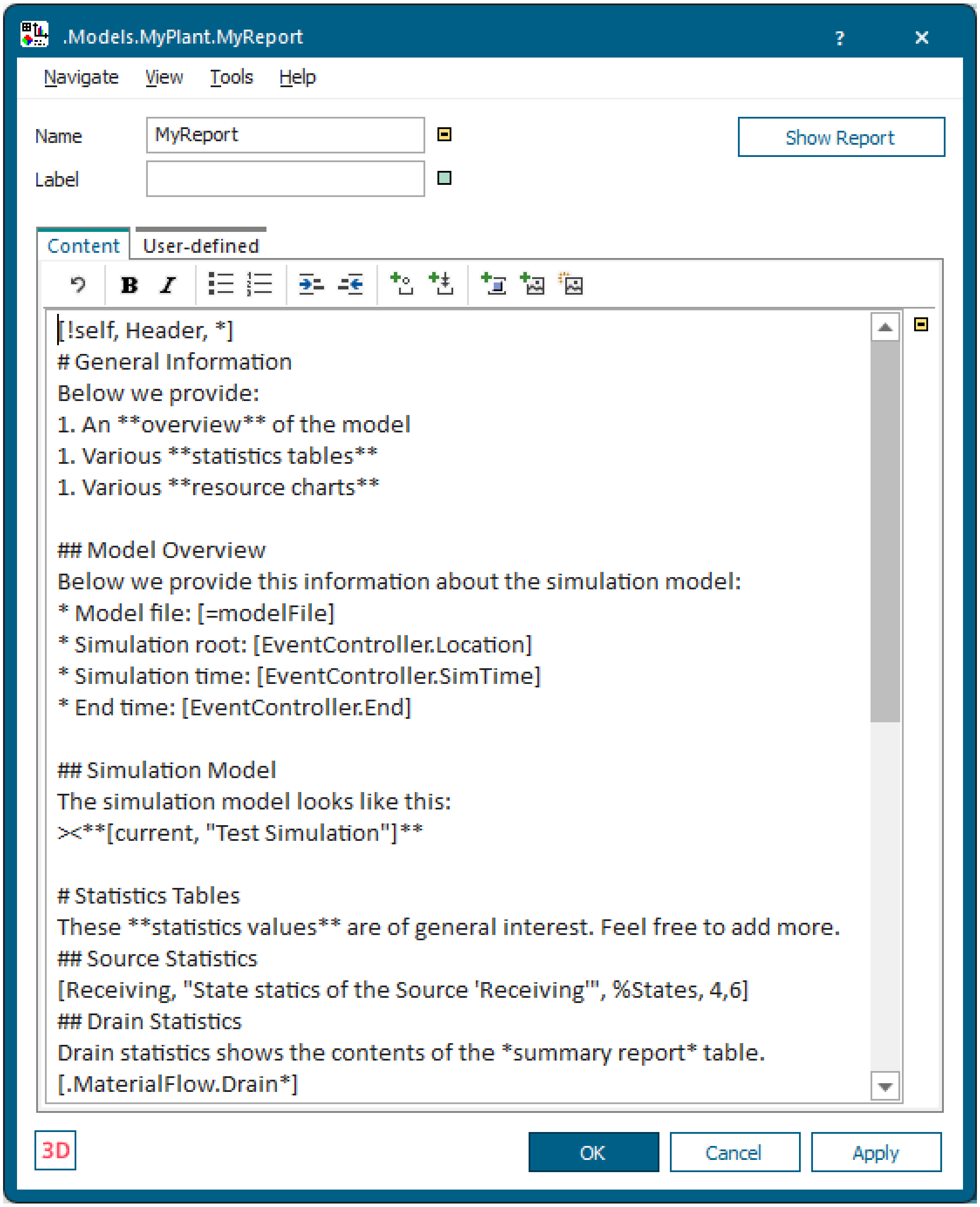
See also

Tab Content [HtmlReport]

Tab Content

Tab Content [HtmlReport]

Define the content which the *HtmlReport* shows on the tab **Content**.



Remarks

The built-in report provides examples of the most important items that you might use for your standard reports.

The check box to the right of the canvas turns inheritance off or on.

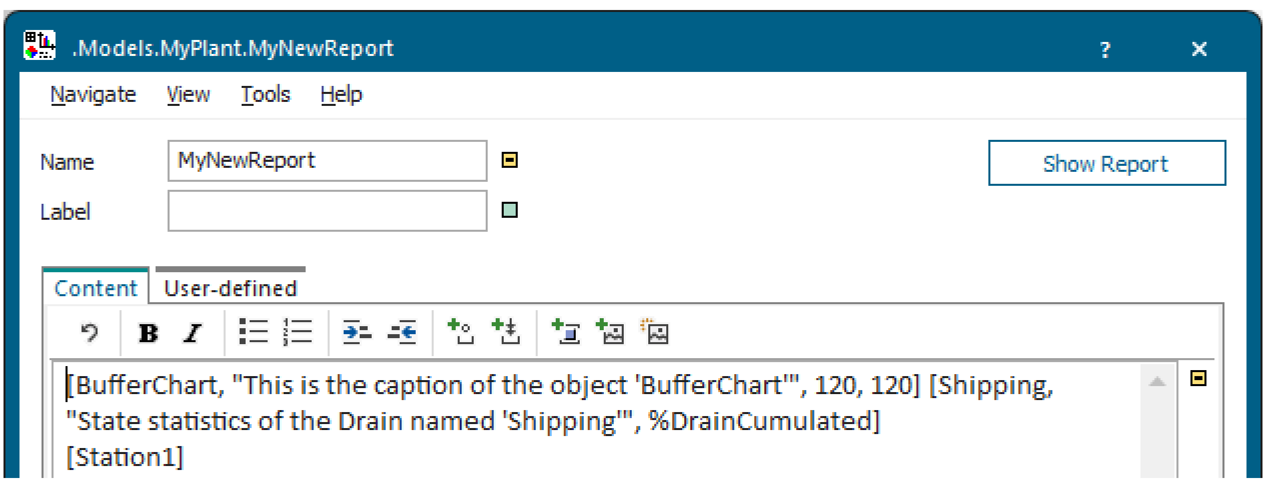
- A green check box means that inheritance is turned on. Then the object inherits, i.e., uses, the value from the parent object from which it was derived. When you change the value of the parent object, *Plant Simulation* also changes the value of the child object.
- An orange check box with a minus means that inheritance is turned off. Values you type in only apply to the selected object.

You can **type in** the content of the *HtmlReport* as **text**, or you can **type in and select settings** in the dialog **Object Parameters**. You define special elements, such as headings, tables, and pictures in **text form**. For this purpose the *HtmlReport* provides a markup language which is designed for easy readability. The syntax is a superset of HTML, i.e., you can also type in the text in HTML syntax. We advise against doing this as HTML syntax is complicated to type and hard to read. You can also **drag** an object from the *Frame* to the tab **Content and drop** it there.

You can also **drag objects** from the *Frame* onto the icon of the *HtmlReport* **and drop them** there.

- If the **Content** of the *HtmlReport* is inherited, the data of the objects replaces the pre-defined content of the *HtmlReport*. *Plant Simulation* opens the display window of the *HtmlReport* automatically.
- If the **Content** of the *HtmlReport* is not inherited, *Plant Simulation* appends the data of the objects to the pre-defined content of the *HtmlReport*. *Plant Simulation* opens the display window of the *HtmlReport* automatically.

To open the dialog **Object Parameters** for an object, which you added to the tab **Content** with **drag-and-drop**, click the mouse between the name of the object and the closing bracket, and click .



Then, select the settings you need in the dialog **Select Object**.

You can:

- Use the **Default Settings of the HtmlReport**
- **Define a Heading**

- Create a Bulleted List
- Create a Numbered List
- Highlight Text by Bolding and/or Italicizing It
- Indent Text
- Center Text
- Add a Reference to an Object on the HTML Page
- Add an Image of an Object
- Add a Bitmap File
- Collect Object Instances and Show Them
- Insert a Comment into Your Report
- Insert a Horizontal Line
- Insert HTML Tags
- Protect Characters
- Use a String Unchanged in the HTML Code

You can add some of the formatting options by clicking the respective button on the **Toolbar of the HtmlReport** or by pressing a key combination..

See also

Show Statistics and Other Values in a Report

Show Reports in a Hierarchically Structured Model

Toolbar of the HtmlReport

You can add some of the formatting options by clicking the respective button on the toolbar or by pressing a key combination.

Remarks

The toolbar of the *HtmlReport* provides these buttons.

To do this	Click	Key combination
Undo your last action		**Ctrl+Z**
Apply bold face to the selected text		**Ctrl+B**
Italicize the selected text		**Ctrl+I**
Convert the selected text to a numbered list		—
Convert the selected text to a bulleted list		—
Increase the indent of the selected text by one indent level		**Ctrl+M**
Decrease the indent of the selected text by one indent level		**Ctrl+Shift+M**
Insert a nonbreaking space (nonbreaking spaces are not wrapped and double-nonbreaking spaces are not removed)		**Ctrl+Shift+Hyphen (-)**
Insert a page break, which is only visible in the printed report		**Ctrl+Shift+Enter**
Add a Reference to an Object		**Ctrl+K**
Add an Image of an Object		—
Add a Bitmap File		—
Spell Check the Content of the HtmlReport		**Ctrl+S**

You can press these buttons to insert markups.

To do this	Click	Key combination
Insert the markup for a heading of level 1 or remove it again	—	**Ctrl+1**
Insert the markup for a heading of level 2 or remove it again	—	**Ctrl+2**
Insert the markup for a heading of level 3 or remove it again	—	**Ctrl+3**
Insert the markup for a heading of level 4 or remove it again	—	**Ctrl+4**
Insert the markup for a heading of level 5 or remove it again	—	**Ctrl+5**
Insert the markup for a heading of level 6 or remove it again	—	**Ctrl+6**
Remove all markups for headings, no matter what level they are	—	**Ctrl+0**
Insert the markup for centering a line	—	**Ctrl+E**
Insert a soft hyphen/ syllable hyphen to allow hyphenating words in the *HtmlReport*	—	**Ctrl+-**

Default Settings of the HtmlReport

We defined a number of default settings for the *HtmlReport*.

You can adapt them to your needs with the options described under the **Tab Content**.

Drain Statistics

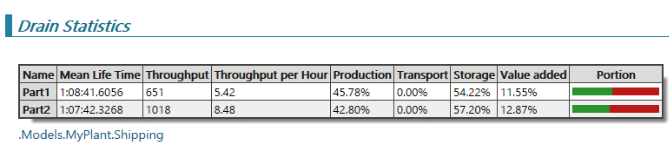
The setting `## Drain Statistics` shows these values by default.

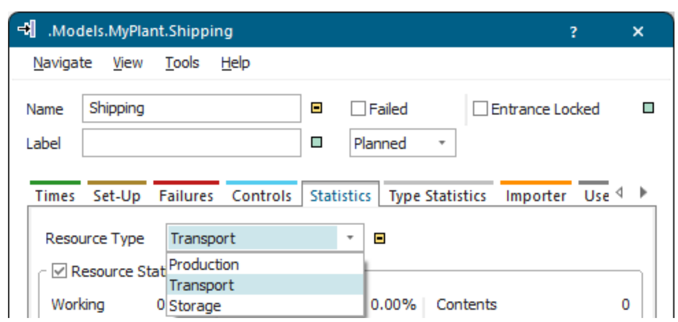
Item English	Description	Read-Only Attribute	Item German
Object	Shows the names of all *Drain* objects within the entire simulation model that removed parts.	—	**Objekt**
Name	Shows the names of the part types which the *Drain* in this row removed from the plant.	—	**Name**
Mean Life Time	Shows the mean life time (throughput time) of all parts of this part type which the *Drain* in this row removed from the plant.	**StatAvgLifeSpan [SimTalk] - Drain**	**Mittlere Durchlauf zeit**
Total Through put	Shows the total throughput of all parts of this part type which the *Drain* in this row removed from the plant.	**StatDeleted [SimTalk] - Drain**	**Durchsatz**
Through put per Hour	Shows the number of parts that the *Drain* removed from the plant in an hour over all observed times during which the *Drain* was available.	**StatThroughput PerHour [SimTalk] - Drain**	**Durchsatz pro Stunde**

Item English	Description	Read-Only Attribute	Item German
Production	Shows the time portion during which the parts spent on a resource object of type **production**.	—	**Produktion**
Transport	Shows the time portion during which the parts spent on a resource object of type **transport**.	—	**Transport**
Storage	Shows the time portion during which the parts spent on a resource object of type **storage**.	—	**Lagerung**
Value Added	Shows the time portion during which the parts were processed and were located on resource objects of type **production**. This is the time during which the value of the parts was increased.	StatProdWorkingPortion [SimTalk] - Drain	**Wertschöpfung**
Portion	Shows the accumulated values of the columns **production** (green), **transport** (orange), and **storage** (red) as bar segments.	—	**Anteil**

Colors for the Portion

The colored bars for the **Portion** show the accumulated values of the columns **Production** (green), **Transport** (orange), and **Storage** (red) as bar segments.





Define a Heading

Type in a number sign # at the start of a line to create a heading of level 1.

Remarks

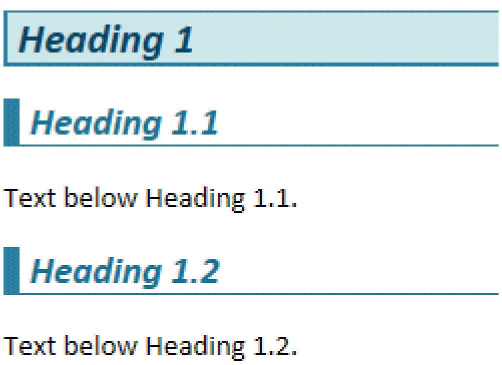
Type in two number signs ## to create a heading of level 2.

You can specify a maximum of four heading levels, i.e., you can type in #####.

The *HtmlReport* also shows the headings of level 1 and 2 in the table of contents in the structure pane on the left-hand side of the HTML page.

```
# Heading 1
## Heading 1.1
Text below Heading 1.1.
## Heading 1.2
Text below Heading 1.2.
```

The result looks like this:



See also

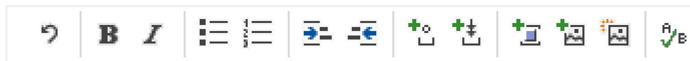
Show Statistics and Other Values in a Report

Create a Bulleted List

Type in an asterisk followed by a space * at the start of a line to create a bulleted list.

Remarks

Each asterisk followed by a space at the start of a line creates a new bullet. Each bullet can span several lines. Terminate the bulleted list with an empty line.

Instead, you can also click  on the toolbar .

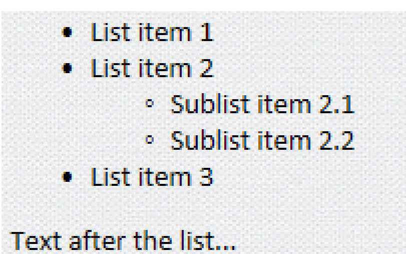
Insert a tab stop or type in four spaces at the start of the line, followed by * to create a sublist.

Each additional four empty spaces or each additional tab stop create sub-sublists which are indented further to the right.

```
* List item 1
* List item 2
    * Sublist item 2.1
    * Sublist item 2.2
* List item 3

Text after the list...
```

The result looks like this:



See also

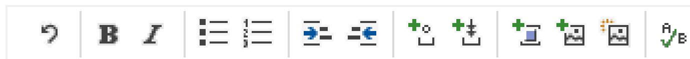
Show Statistics and Other Values in a Report

Create a Numbered List

Type in `1.` followed by a space `1. ` at the start of a line to create a numbered list.

Remarks

Type in `1. ` at the start of a line creates a new numbered item in the list. Instead of a `1`, you can also type in any other number in front of the numbered item. *Plant Simulation* ignores the number and always creates the correct numbering which starts with 1. Each numbered item can span several lines. Terminate the numbered list with an empty line.

Instead, you can also click  on the toolbar .

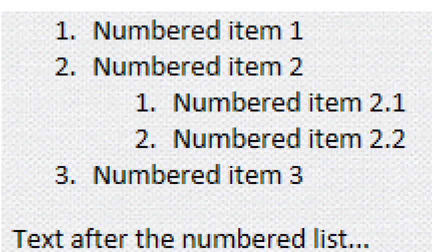
Insert a tab stop or type in four spaces at the start of the line, followed by `1. ` to create a sublist within the numbered list.

Each additional four spaces or each additional tab stop create sub-sublists within the numbered list which are indented further to the right.

```
1. Numbered item 1
1. Numbered item 2
    1. Numbered item 2.1
    1. Numbered item 2.2
1. Numbered item 3

Text after the numbered list...
```

The result looks like this:


1. Numbered item 1
2. Numbered item 2
    1. Numbered item 2.1
    2. Numbered item 2.2
3. Numbered item 3

Text after the numbered list...

Note

You can also create bulleted sublists within numbered lists.

```
1. Numbered item 1
1. Numbered item 2
        * Sublist item 2.1
        * Sublist item 2.2
1. Numbered item 3
```

```
Text after the numbered list...
```

1. Numbered item 1
2. Numbered item 2
    - Sublist item 2.1
    - Sublist item 2.2
3. Numbered item 3

Text after the numbered list...

See also

Show Statistics and Other Values in a Report

Highlight Text by Bolding and/or Italicizing It

Type in a space followed by an asterisk `*`, followed by the `text`, which you would like to show, and then an asterisk followed by a space `*`, to italicize the enclosed text.

Remarks

Instead of an empty space, you can also use other spaces, such as a tab stop or a line break. The enclosed text cannot start or end with a space so that italicized text can be distinguished from a bulleted list (* at the start of a line) and from the multiplication sign (**2*3** or **2 * 3**).

The italicized text range can span several lines and paragraphs.

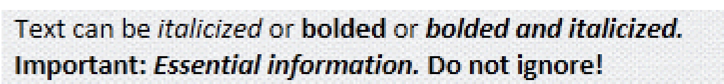
Enclose the text between `**` and `**` to highlight this text in **bold face**.

Enclose the text between `***` and `***` to highlight this text in **bold face** and to *italicize* it.

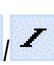
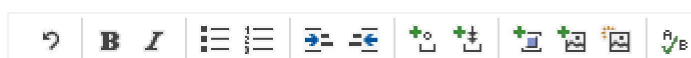
You can also increase highlighting consecutively: If you use * in already *italicized* text, it will be displayed in **bold face** in addition. If your text is already **bolded**, and you use *, it will be displayed in **bold face** and will be *italicized*.

```
Text can be *italicized* or **bolded** or ***bolded and italicized.***
**Important: *Essential information.* Do not ignore!**
```

The result looks like this:


Text can be *italicized* or **bolded** or ***bolded and italicized.***
Important: *Essential information.* Do not ignore!

> **Note**
>
> *Plant Simulation* does not interpret the character combination space asterisk * as the start of *italicized* text to prevent that object references, such as ***.MUs.Part:1**, are going to be misinterpreted.

Instead, you can also click  /  on the toolbar .

See also

Show Statistics and Other Values in a Report

Indent Text

Type in a greater than sign > at the start of a line to indent a paragraph.

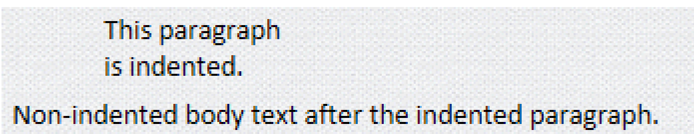
Remarks

A paragraph can span several lines. Terminate it with an empty line. To indent the paragraph more, type in more than one greater than sign >> at the start of the line.

```
> This paragraph
is indented.

Non-indented body text after the indented paragraph.
```

The result looks like this:



See also

Show Statistics and Other Values in a Report

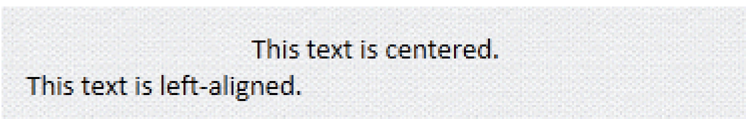
Center Text

Type in a greater than sign followed by a less-than sign >< at the start of a line to center the text in this line.

Remarks

```
>< This text is centered.
This text is left-aligned.
```

The result looks like this:



See also

[Show Statistics and Other Values in a Report]

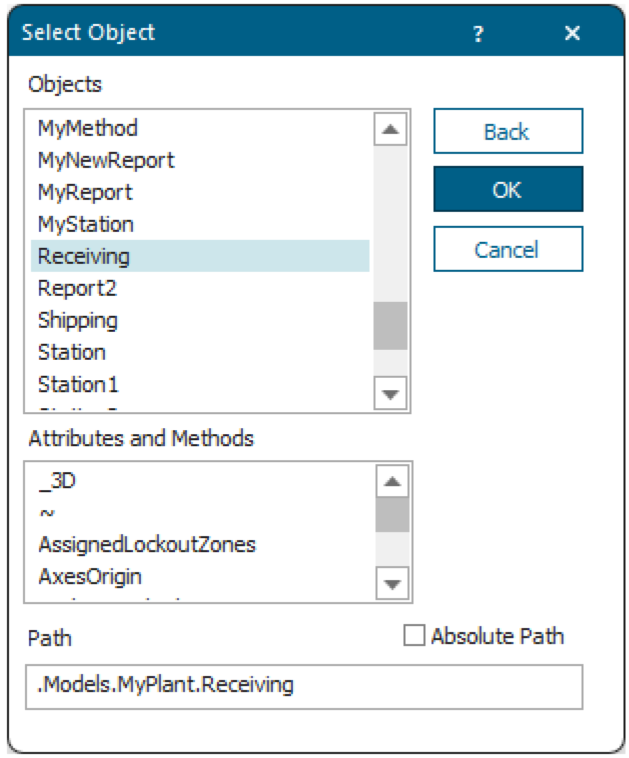
Add a Reference to an Object

Add a Reference to an Object on the HTML Page

Specify the settings for displaying a *reference to an object* on the HTML page.

Type Object Parameters into the Dialog Object Parameters

To select an object or an object attribute to be displayed on the HTML page, click  on the toolbar . Then select the object and/or the value of an attribute or a method of this object, which you would like to show, in the dialog **Select Object**. Click **OK**.

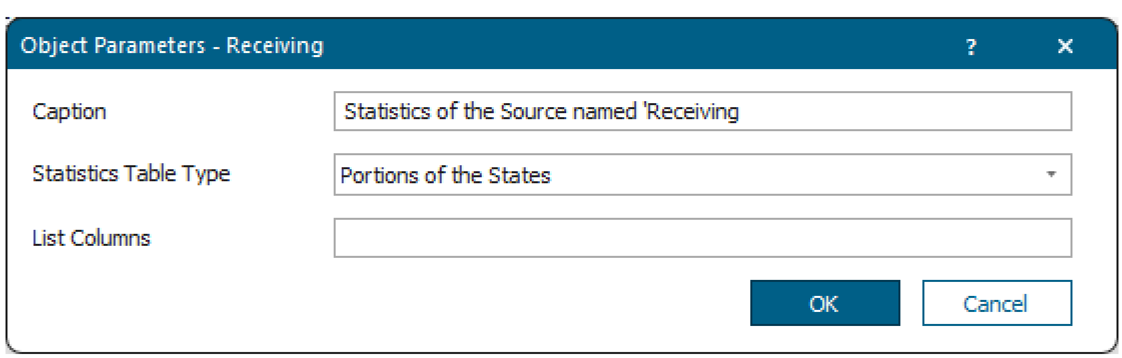


In this case the path, which was detected within brackets when you held down **Shift** while clicking **Add Object** is preselected.

Type in the **Caption** which the *HtmlReport* shows below the statistics table on the HTML page.

Select the **Statistics table type** you want to show for the object. The different object types provide different statistics table types.

If the text cursor is located within an object reference ([...]), *Plant Simulation* opens the dialog **Select Object Parameters**. In this case the object parameter is preselected that was detected within brackets when you clicked **Add Object** .



You can:

- Display an Attribute of an Object
- Display an AttributeExplorer
- Display a Broker
- Display a Chart
- Display a Checkbox
- Display an Object of Type Comment
- Display a DataTable or a DataList
- Display a Drain
- Display an Object of Type Display
- Display a Dropdown List
- Display a FileLink
- Compute a Formula
- Display a Frame
- Display a GanttChart
- Display a HtmlReport
- Display an Icon of an Object
- Display a Method
- Display a Picture of a Scene
- Display a PythonModule
- Display a SankeyDiagram
- Display Statistics Values of an Object as a Table
- Display a Table from the SQLite Interface
- Display a Variable

Type in the Parameters on the Tab Content

You also type in the instructions that correspond to the settings you can type into the dialog.

Type in an opening bracket, followed by the name of the object, followed by the closing bracket `[ObjectName]` to access that object, to read attribute values of this object, to use its icon, or to display the object itself on the HTML page.

To display the **value of an attribute** on the HTML page, you can also type in the path to the attribute in brackets. Relative paths are resolved starting with the *Frame* in which the *HtmlReport* is located. To read the attributes of the *HtmlReport* itself, use the identifier `self` to access it.

Depending on the type of object, you can add additional arguments to fine-tune the HTML output. You can either add arguments in a specific order separated by commas or by name separated by white spaces.

Compare these examples, which produce the same output.

```
[.Models.Model._3D, "caption", 100%, %planningview]
```

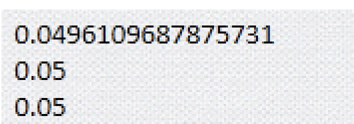
```
[.Models.Model._3D Caption="caption" ScaleTo=100% PlanningView=true]
```

If the attribute contains a floating point value, you can round the value by typing in the number of decimal places, separated by a comma.

To force a certain number of decimal places, even when these are zeros, type a `0` in front of the number of decimal places.

```
[Station1.statFailPortion]
[Station1.statFailPortion, 2]  // round to 2 digits
[Station1.statFailPortion, 02] // 2 decimal places
```

The result looks like this:

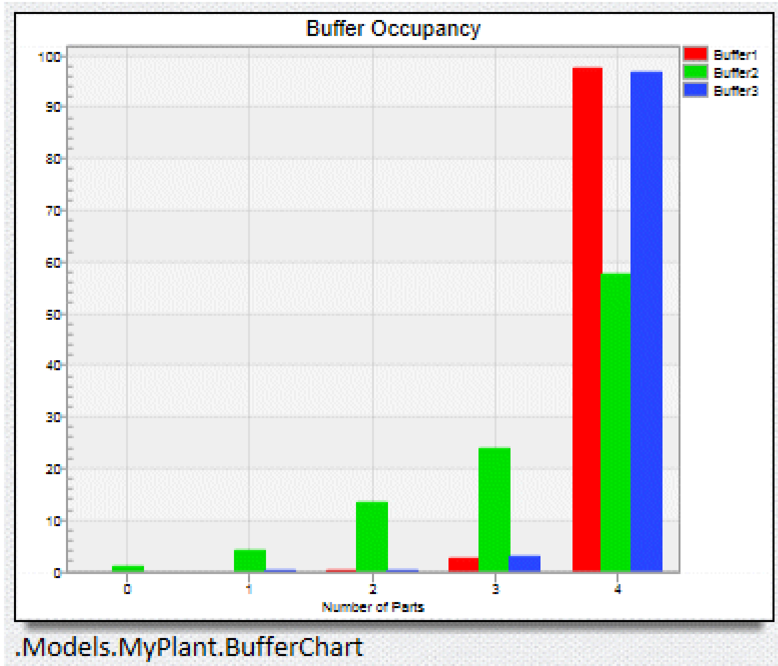


Type in * to Display all Instances of an Object

Instead of displaying a single object on an HTML page by typing in its name within brackets, you can also display all instances of an object one after the other.

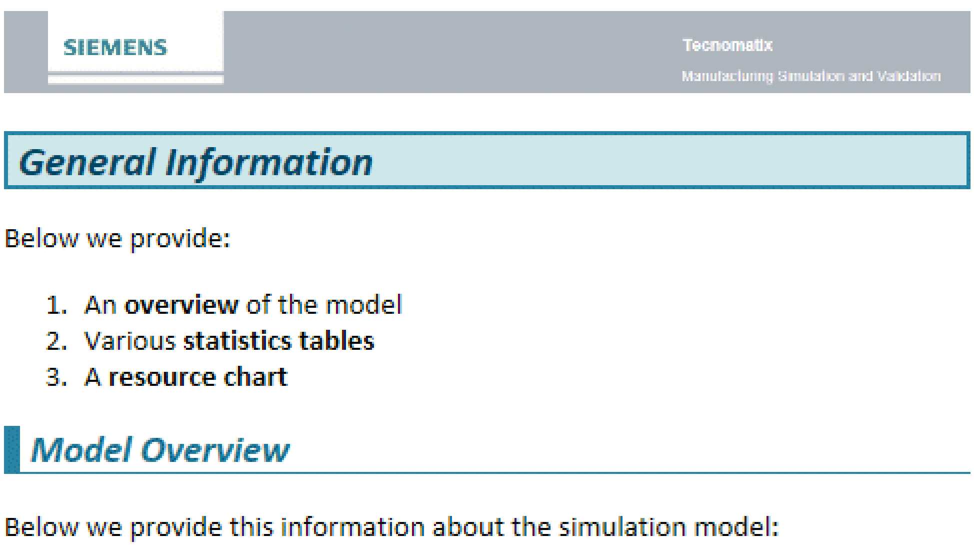
To do so, type in an asterisk * after the path to the object. *Plant Simulation* then displays all instances which inherit directly or indirectly from the object and which are located in the *Frame* into which you inserted the *HtmlReport*. This includes *sub-Frames*.

The result looks like this:




```
[.UserInterface.HtmlReport*]     // gathers all HtmlReports inserted into the model
```

The result looks like this:



If you select a *Frame* to be inserted, which provides the *user-defined attribute* named *StatisticsObject*, the dialog **Object Parameters** shows the available statistics functions.

If you create a *user-defined attribute* named *ReportObject* for an object, *Plant Simulation* does not display this object but the *ReportObject* instead. If the *user-defined attribute* has the data type *object*, *Plant Simulation* displays the object which the attribute references. In many cases this will be an *HtmlReport* which inserts its report into the HTML page. If you want to insert a *DataTable* or the result of a method call, you do not have to reference an object of type *DataTable* or *Method*, but you can set the data type of the *user-defined attribute* to *table* or *method* and define the table or method directly in the attribute.

If you want to reference another object in a *Frame*, you do not have to create a *user-defined attribute* of data type *object*, but you can rename an inserted object to *ReportObject*. This object will then be displayed on the HTML page instead of the *Frame*.

See also

Show Statistics and Other Values in a Report

Show Reports in a Hierarchically Structured Model

Display an Attribute of an Object

Display attributes with a certain precision.

Specify Named Arguments

For the data types *Real*, *Length*, *Weight*, *Speed*, *Time*, and *Acceleration* the named argument *Precision* specifies the **Integer Places** and/or the **Decimal Places** with which the attribute is shown in the *HtmlReport*.

The example shows two decimal places of our user-defined attribute named a of the *Station*.

```
[Station.a precision=2]
```

Display an AttributeExplorer

Specify the settings for displaying an *AttributeExplorer* on the HTML page.

Select the Object in the Dialog Select Object

To select an **AttributeExplorer** whose **attribute table** you want to show on the HTML page, click on the toolbar . Then select the *AttributeExplorer* in the dialog **Select Object** and click **OK**.

Type in the Parameters on the Tab Content

You can type in the following instructions to set the columns to be displayed.

```
[AttributeExplorer]
[AttributeExplorer, "Tab. 1: Mean value of the throughput times"]
[AttributeExplorer, 0, 2..4, 7]
[AttributeExplorer, *..4, 7]
[AttributeExplorer, 0, 2..4, 7..*]
[AttributeExplorer, 0, 2..4, 7, #ColumnName]
```

The result looks like this:

	ProcTime	ExitStrategy	FailImporterActive	FailureActive	Label	Failures.Failure.MTTR	XDim	YDim
Drain	0.0000		false	true	Shipping			
MyStation	1:00.0000	Cyclic	false	true	My Schärer Machine	1:00.0000		
ParallelStation	1:00.0000	Cyclic	false	true	Multilathe		2	2
Source		Cyclic	false	true	Receiving			
Machine	1:00.0000	Cyclic	false	true				
MyMachine	1:00.0000	Cyclic	false	true				
MyStation	1:00.0000	Cyclic	false	true	My Schärer Machine	1:00.0000		
Station1	1:00.0000	Cyclic	false	true				

	ProcTime	ExitStrategy	FailImporterActive	FailureActive	Label	Failures.Failure.MTTR	XDim	YDim
Drain	0.0000		false	true	Shipping			
MyStation	1:00.0000	Cyclic	false	true	My Schärer Machine	1:00.0000		
ParallelStation	1:00.0000	Cyclic	false	true	Multilathe		2	2
Source		Cyclic	false	true	Receiving			
Machine	1:00.0000	Cyclic	false	true				
MyMachine	1:00.0000	Cyclic	false	true				
MyStation	1:00.0000	Cyclic	false	true	My Schärer Machine	1:00.0000		
Station1	1:00.0000	Cyclic	false	true				

Tab. 1: Mean value of the throughput times

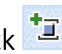
See also

Show Statistics and Other Values in a Report

Display a Broker

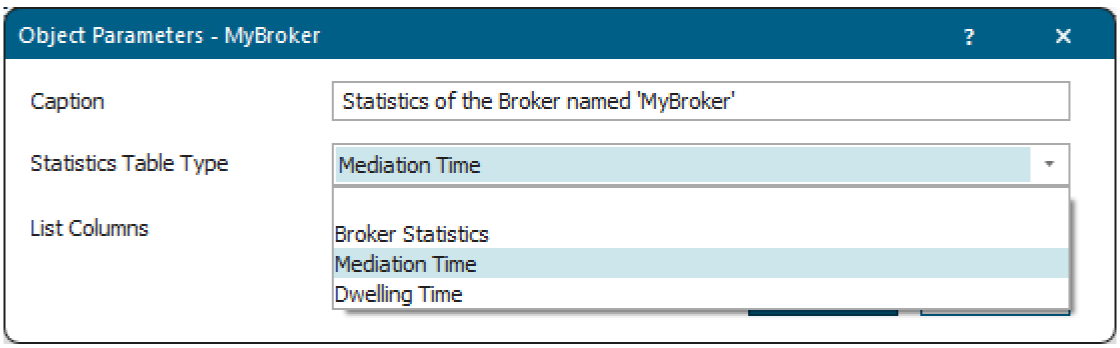
Specify the settings for displaying a *Broker* on the HTML page.

Type Object Parameters into the Dialog Object Parameters

To display the **statistics table** of the **Broker** on the HTML page, click  on the toolbar . Then select the *Broker* in the dialog **Select Object** and click **OK**.

Type in the **Caption** which the *HtmlReport* shows below the statistics table of the *Broker*.

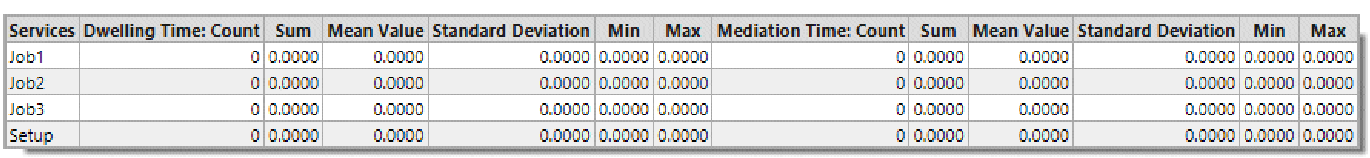
Select the **Statistics table type** you want to show for the *Broker*.



If you do not select the **Statistics Type Table**, the *HtmlReport* displays the **broker statistics table**.

```
[MyBroker] // our Broker is called MyBroker
[MyBroker, "Tab. 1: Broker Service Statistics"]
[Broker1, 0, 2..4, 7]
[Broker2, *..4, 7]
[Broker3, 0, 2..4, 7..*]
```

The result looks like this:



Services	Dwelling Time: Count	Sum	Mean Value	Standard Deviation	Min	Max	Mediation Time: Count	Sum	Mean Value	Standard Deviation	Min	Max
Job1	0	0.0000	0.0000	0.0000	0.0000	0.0000	0	0.0000	0.0000	0.0000	0.0000	0.0000
Job2	0	0.0000	0.0000	0.0000	0.0000	0.0000	0	0.0000	0.0000	0.0000	0.0000	0.0000
Job3	0	0.0000	0.0000	0.0000	0.0000	0.0000	0	0.0000	0.0000	0.0000	0.0000	0.0000
Setup	0	0.0000	0.0000	0.0000	0.0000	0.0000	0	0.0000	0.0000	0.0000	0.0000	0.0000

Tab. 1: Broker Service Statistics

See also

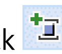
Display Statistics Values of an Object as a Table

Show Statistics and Other Values in a Report

Display a Chart

Specify the settings for displaying a *Chart* on the HTML page.

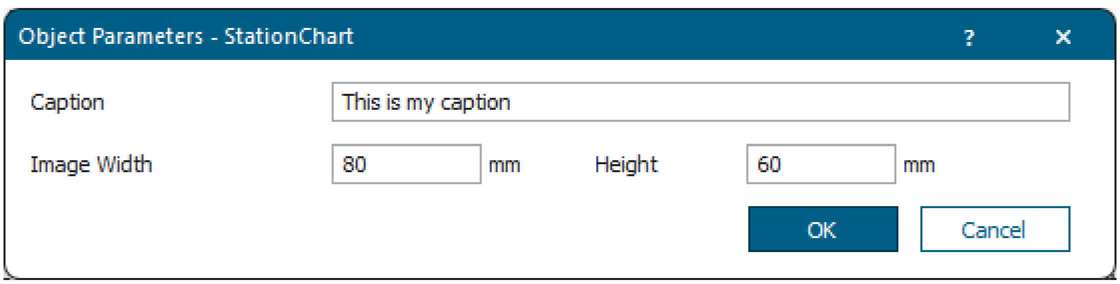
Type Object Parameters into the Dialog Object Parameters

To select an **Chart** whose **diagram picture** you want to show on the HTML page, click  on the toolbar . Then select the *Chart* in the dialog **Select Object** and **OK**.

Type in the **Caption** which the *HtmlReport* shows below the *Chart*.

Type in the **Image Width** with which the *HtmlReport* shows the picture of the object. You can also type in an asterisk (*) for either the **Width** or the **Height** to ensure that the image will not be distorted.

Type in the **Image Height** with which the *HtmlReport* shows the picture of the object. You can also type in an asterisk (*) for either the **Width** or the **Height** to ensure that the image will not be distorted.

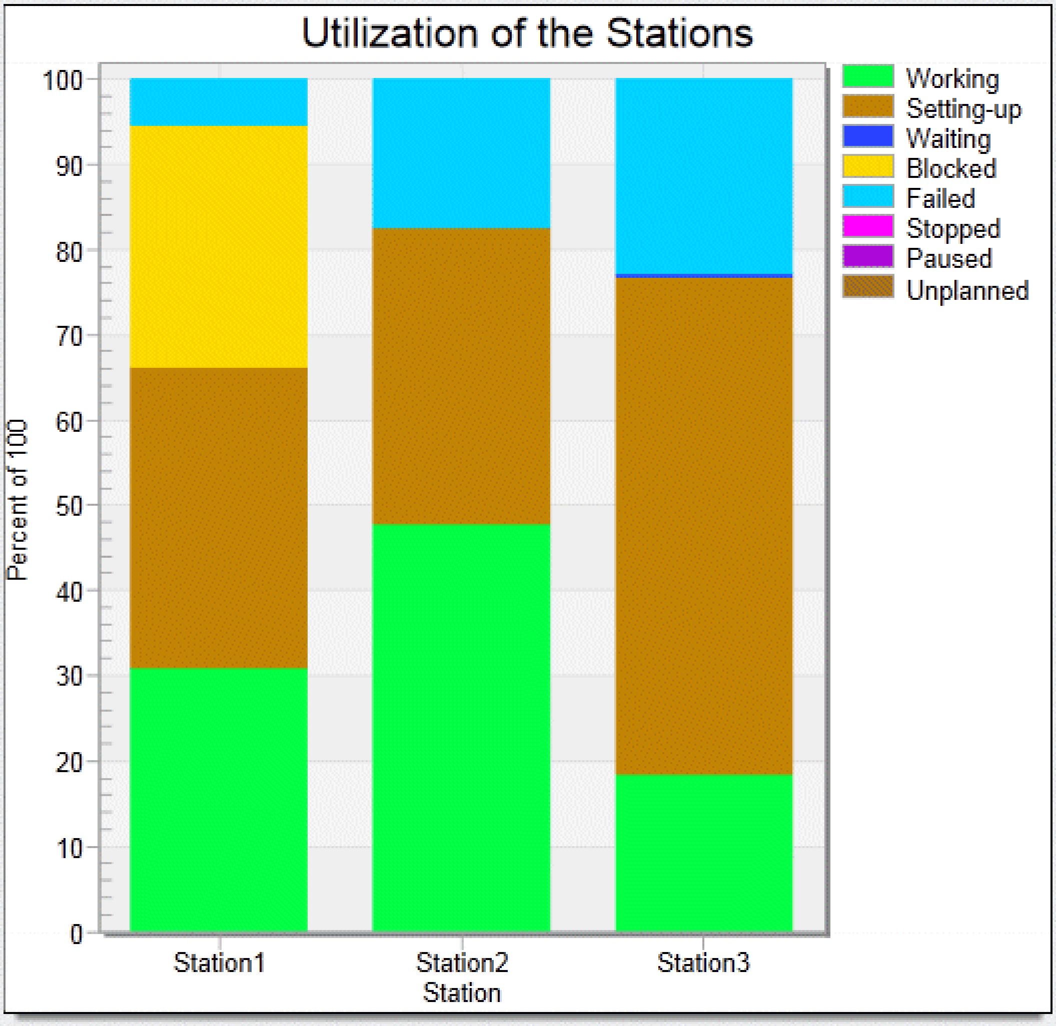


Type in the Parameters on the Tab Content

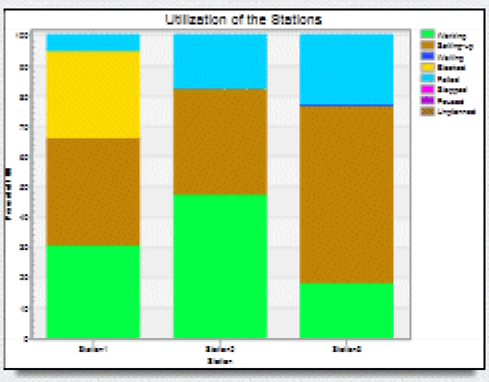
Instead, you can also type in the instructions that correspond to the settings you can enter into the dialog.

```
[StationChart] // our Chart is called StationChart
[StationChart, 80, 60]
[StationChart, "This is my caption"]
[StationChart, "This is my caption", 80, 60]
```

The result looks like this:



This is my caption



This is my caption

Specify Named Arguments

You can specify the following named arguments:

- The named argument *Caption* specifies the caption with which the specified *Chart* is shown in the *HtmlReport*.

  ```
  [Chart Caption="This is my caption"]
  ```

- The named argument *Size* specifies the size with which the specified *Chart* is shown in the *HtmlReport*.

  You can specify the size like this:

  - Do not specify the argument to show the picture of the *Chart* with its default size.
  - Specify `[*, y]` to set the height of the picture of the *Chart* to y and to use the width of the default width of the picture.
  - Specify `[x, *]` to set the width of the picture of the *Chart* to x and to use the height of the default height of the picture.
  - Specify `[x, y]` to scale the picture of the *Chart* to the size x and the height y. This might distort the picture though.

  ```
  [Chart Size=80,60]
  ```

See also

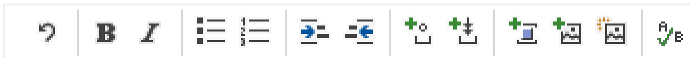
Show Statistics and Other Values in a Report

Show Reports in a Hierarchically Structured Model

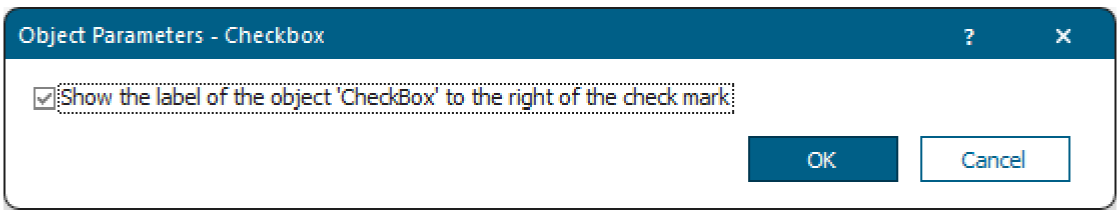
Display a Checkbox

Specify the settings for displaying a *Checkbox* on the HTML page.

Type Object Parameters in the Dialog Object Parameters

To display a **Checkbox**, which you inserted into your simulation model, on the HTML page, click on the toolbar . Then select the *Checkbox* in the dialog **Select Object** and click **OK**

To show the **Label** of the object *Checkbox* to the right of the check mark on the HTML page, select the check box.



To display the **Label** of the check box to the right of the icon ✓, you can also type in the following:

```
[Checkbox] or
[Checkbox, true]
```

The result looks like this:

✓ Enter your text here

To display the **Label** of the check box to the left of the icon ✓, type in:

```
[Checkbox, false]
```

The result looks like this:

Enter your text here ✓

Specify Named Arguments

The named argument *CheckBeforeText* specifies if the check mark of the *Checkbox* is shown to the left of the text in the *HtmlReport* (`true`) or if it is shown to the right (`false`).

The default value is false.

```
[Checkbox CheckBeforeText=true]
```

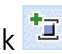
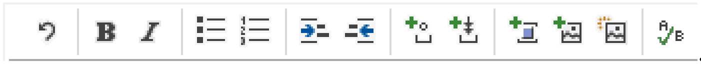
See also

Show Statistics and Other Values in a Report

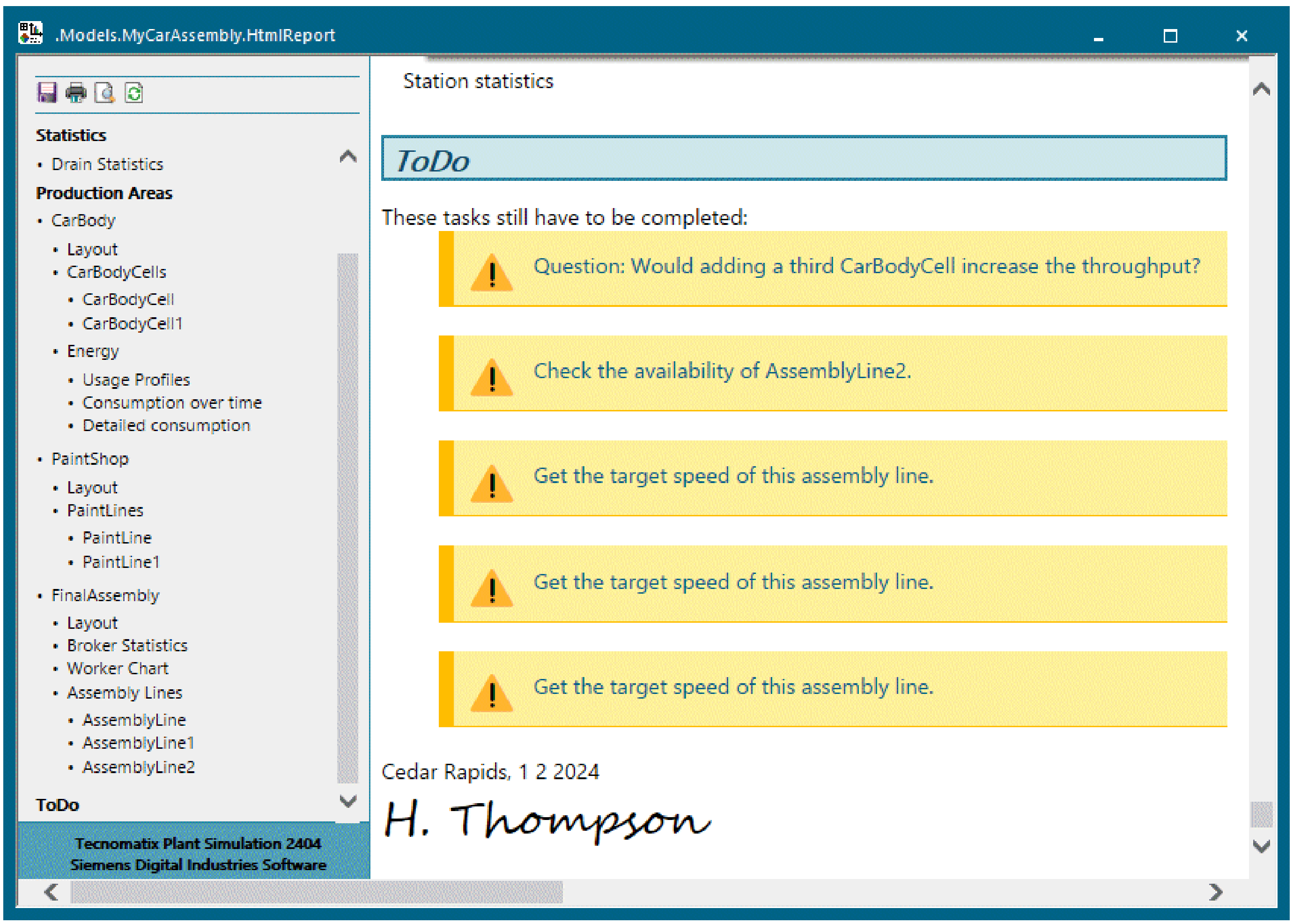
Display an Object of Type Comment

Specify the settings for displaying an object of type *Comment* on the HTML page.

Select the Object in the Dialog Select Object

To display a **Comment**, which you inserted into your simulation model, on the HTML page, click  on the toolbar . Then select the *Comment* in the dialog **Select Object** and click **OK**.

The *HtmlReport* shows the text you typed in on the tab **Comment** to draw attention to information that you deem important. When you click the text/link, *Plant Simulation* opens the *Comment* object on the tab **Comment**.



Specify the Parameters on the Tab Content

Instead of selecting the object, you can also type it in directly:

```
[Comment]
```

See also

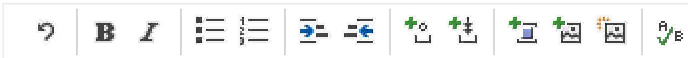
Show Statistics and Other Values in a Report

Show Reports in a Hierarchically Structured Model

Display a DataTable or a DataList

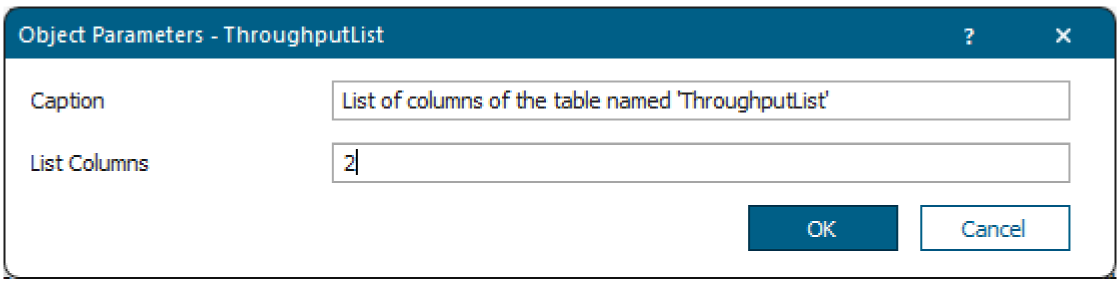
Specify the settings for displaying a *DataTable/list object* on the HTML page.

Type Object Parameters in the Dialog Object Parameters

To display a **DataTable** or a list object (**DataList**, **DataQueue**, **DataStack**) in table form on the HTML page, click  on the toolbar . Then select the *DataTable/list object* in the dialog **Select Object** and click **OK**.

Type in the **Caption** which the *HtmlReport* shows below the *DataTable/List*.

Type in the identifiers of the **List columns** of the *DataTable/list object* which the *HtmlReport* shows on the HTML page.



Type the Parameters on the Tab Content

Instead, you can also type in the instructions that correspond to the settings you can type into the dialog.

```
[ThroughputList] // our DataTable is called ThroughputList
[ThroughputList, "Figure 1: Mean value of the throughput times"]
[ThroughputList, 0, 2]       // columns 0 and 2
[ThroughputList, *..4, 7]    // all columns until column 4 and column 7
[ThroughputList, 0, 2..4, 7..*]
[ThroughputList, 0, 2..4, 7, #MyColumnName] // columns 0, 2 to 4, and
column 7, and the columns named MyColumnName
```

The result looks like this:

Part	Total Throughput	Throughput per Hour
Part1	688	5.73
Part2	993	8.28

Figure 1: Mean value of the throughput times

Access Cells of Lists and Tables to Add Their Content to the HtmlReport

The syntax for accessing cells of lists and tables to add their content to the *HtmlReport* is similar to the syntax used in formulas:

```
[MyDataTable[Column, Row]]
```

You can specify the **Column** or **Row** with its number or with its index (if available). The text in the column or row index has to be surrounded by quotation marks and has to be started with the number sign/hash #:

```
[MyDataTable[1, 1]]
[MyDataTable["Name", 1]]
[MyDataTable["Name", "Part type 1"]]
[MyDataTable[#Name, 1]]
[MyDataTable[#Name, #Part type 1]]
```

You have to protect characters, that have a special meaning in the *HtmlReport*, within the text of the column index:

```
[MyDataTable["\"Name\"", 1]
[MyDataTable["Speed\, km/h", 1]]
[MyDataTable[#Speed\, km/h, 1]]
```

Specify Named Arguments

You can specify the following named arguments:

- The named argument *Caption* specifies the caption with which the *DataTable* is shown in the *HtmlReport*.
- The named argument *Columns* specifies the columns of the *DataTable* that are shown in the *HtmlReport*.

  ```
  [DataTable Caption=Test Columns=[1,3,4]]
  ```

See also

Show Statistics and Other Values in a Report

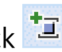
Protect Characters

getHTMLCode [SimTalk] - lists

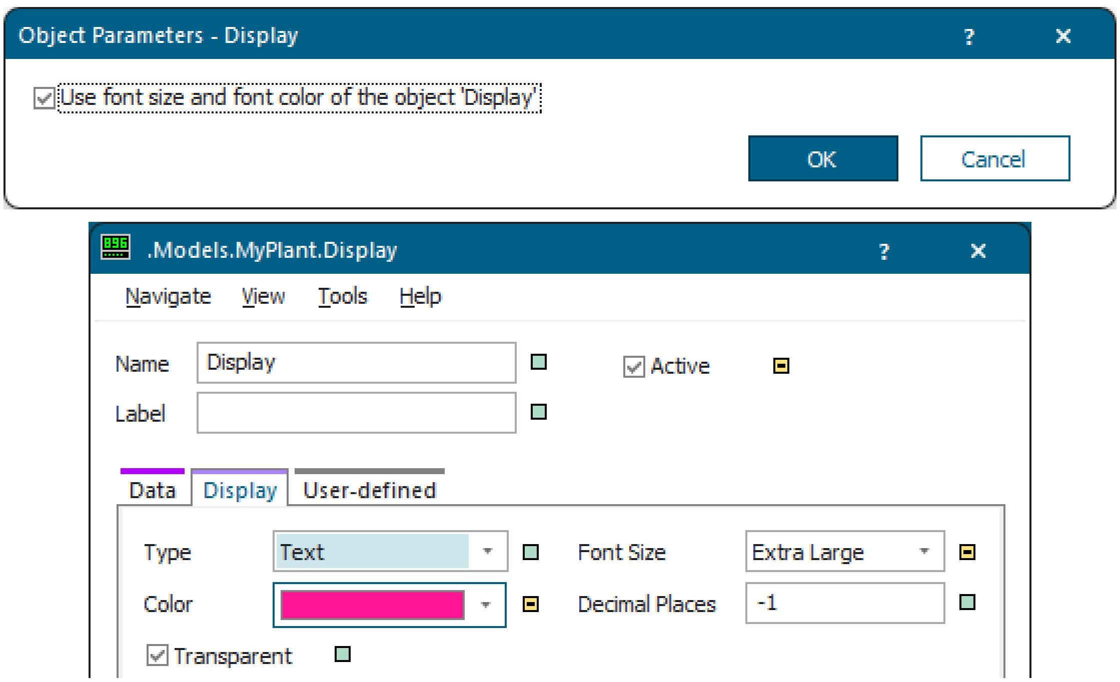
Display an Object of Type Display

Specify the settings for displaying an object of type *Display* on the HTML page.

Type Object Parameters in the Dialog Object Parameters

To display the numeric value of an object of type **Display** on the HTML page, click  on the toolbar . Then select the *Display* in the dialog **Select Object** and click **OK**.

To use the **Font Size** and the **Font Color**, which you selected for the object *Display*, on the HTML page, select 



Type the Parameters on the Tab Content

Instead, you can also type in the instructions that correspond to the settings you can type into the dialog.

You can type in `true` or `false` as the optional parameter. If you type in `true`, *Plant Simulation* displays the value using the **Color** and the **Font Size** which you selected on the tab **Display** of the *Display*.

```
[Display]
[Display, true]
[Display, false]
```

The result looks like this:

Specify Named Arguments

The named argument *UseFontStyle* specifies if the *Display* is shown in the *HtmlReport* with the **Font Size** and the **Font Color** of the *Display* object (`true`) or if it is shown with default values (`false`).

The default value is false.

```
[Display UseFontStyle=true]
```

See also

Show Statistics and Other Values in a Report

Display a Drain

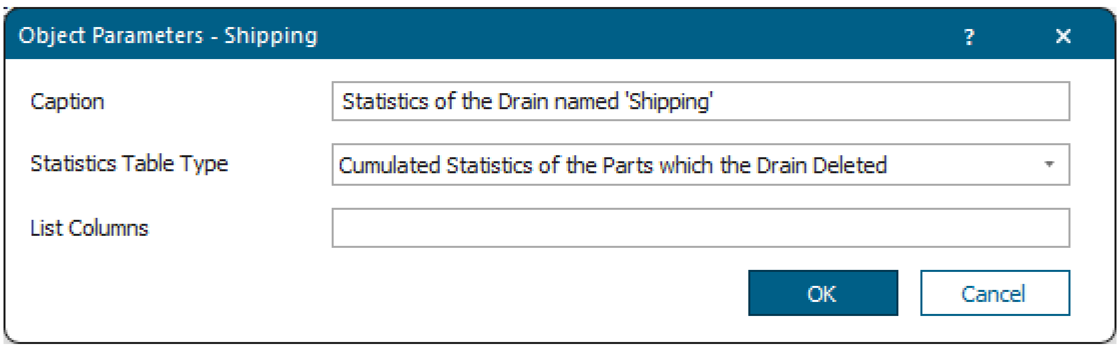
Specify the settings for displaying an object of type *Drain* on the HTML page by specifying object parameters.

Type Object Parameters in the Dialog Object Parameters

To display statistics values of the **Drain** as a table on the HTML page, click  on the toolbar . Then select the *Drain* in the dialog **Select Object** and click **OK**.

Type in the **Caption** which the *HtmlReport* shows below the statistics table of the *Drain*.

Select the **Statistics Table Type** you want to show for the *Drain*.



If you do not select the **Statistics Type Table**, the *HtmlReport* displays the same report on the HTML page that you can activate by checking **Show Summary Report** in the *EventController*. The *Drain report* does not show the first column with the name of the *Drain* as each *Drain* has its own *Drain report* in case no matching statistics table was provided.

Type the Parameters on the Tab Content

Instead, you can also type in the instructions that correspond to the settings you can enter into the dialog.

```
[Shipping, "Statistics of the Drain 'Shipping", %DrainTypesPortions]
```

Shipping	Production							Transport							Storage							
	Working	Set-up	Waiting	Stopped	Failed	Paused	Sum	Working	Set-up	Waiting	Stopped	Failed	Paused	Sum	Working	Set-up	Waiting	Stopped	Failed	Paused	Sum	
Part1	11.55%	9.96%	20.36%	0.00%	3.91%	0.00%	45.78%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	54.22%	0.00%	0.00%	0.00%	54.22%	-
Part2	12.87%	6.47%	20.87%	0.00%	2.59%	0.00%	42.80%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	57.20%	0.00%	0.00%	0.00%	57.20%	-

Statistics of the Drain 'Shipping

See also

Display Statistics Values of an Object as a Table

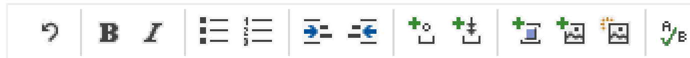
Show Statistics and Other Values in a Report

Show Reports in a Hierarchically Structured Model

Display a Dropdown List

Specify the settings for displaying an object of type *DropDownList* on the HTML page.

Select the Object in the Dialog Select Object

To display the text of the selected item of a **DropDownList**, which you inserted into your simulation model, on the HTML page, click  on the toolbar . Then select the *DropDownList* in the dialog **Select Object** and click **OK**.

Specify the Parameters on the Tab Content

Instead, you can also type in

```
[DropDownList]
```

The result looks like this:

Item 1

See also

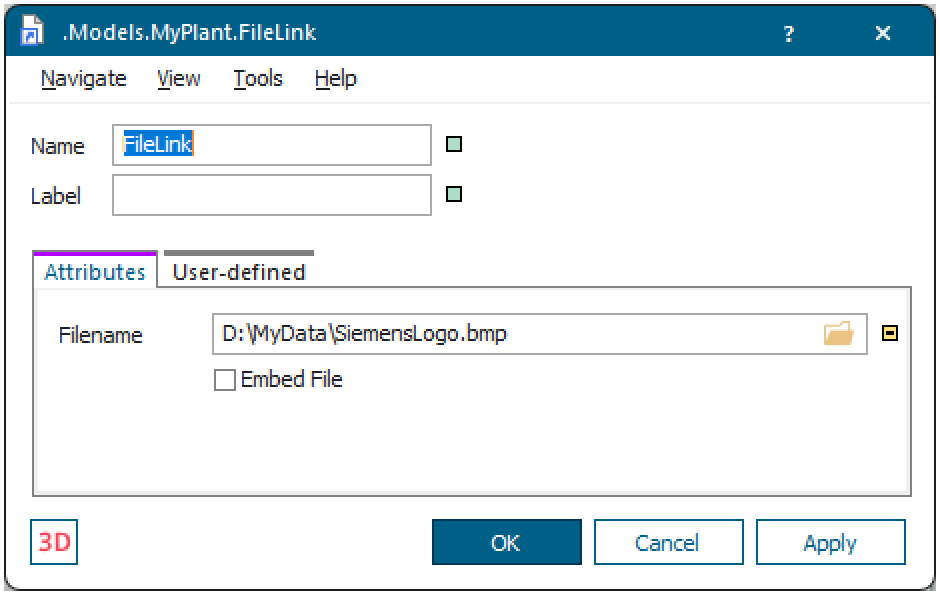
Show Statistics and Other Values in a Report

Display a FileLink

Specify the settings for displaying an object of type *FileLink* on the HTML page.

Type Object Parameters in the Dialog Object Parameters

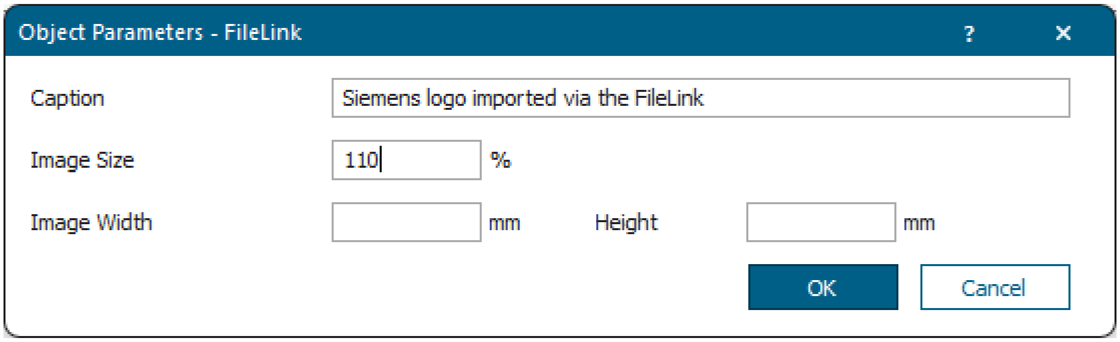
To display a **Bitmap file** or an **SVG file**, which you imported with a **FileLink**, on the HTML page. Then select the **bitmap file** or the **SVG file** in the dialog of the *FileLink* and click **OK**.



Click  on the toolbar . Then select the *DropDownList* in the dialog **Select Object** and click **OK**.

Type in the **Caption** which the *HtmlReport* shows below the *FileLink* image file.

Type in the **Image Size** in percent to which the *HtmlReport* scales the screenshot of the *FileLink*. Instead, you can also enter the **Image Width** and the **Image Height**. You can also type in an asterisk (*) for either the **Width** or the **Height** to ensure that the image will not be distorted. The relative size takes precedence over the absolute size.




```
[FileLink, "Siemens logo imported via the FileLink", 110%]
```

The result looks like this:

SIEMENS
Siemens logo imported via the FileLink

Specify Named Arguments

You can specify the following named arguments:

- The named argument *Caption* specifies the caption with which the specified *FileLink* linked to a picture is shown in the *HtmlReport*.

  ```
  [FileLink Caption="Siemens logo imported via the FileLink"]
  ```

- The named argument *Size* specifies the size of the *icon* (or of the *non-SVG file*).

  You can specify the size like this:

  - Specify `*` or `[*, ]` or do not specify the argument to show the icon with its default size.
  - Specify `[*, y]` to set the height of the icon to y and to use the width of the default width of the icon.
  - Specify `[x, *]` to set the width of the icon to x and to use the height of the default height of the icon.
  - Specify a percentage to scale the icon by the specified factor.
  - Specify `[x, y]` to scale the icon to the size x and the height y. This might distort the icon though.

  The named argument *Size* specifies the size of the *SVG file*.

  You can specify the size like this:
  - Do not specify the argument to show the picture with its default size.

- Specify `[*, y]` to set the height of the picture to y and to use the width of the default width of the picture.
- Specify `[x, *]` to set the width of the picture to x and to use the height of the default height of the picture.
- Specify `[x, y]` to scale the picture to the size x and the height y. This might distort the picture though.

```
[FileLink Size=110%]
```

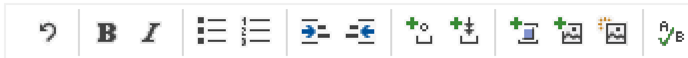
See also

Show Statistics and Other Values in a Report

Display a Frame

Specify the settings for displaying a *Frame* on the HTML page.

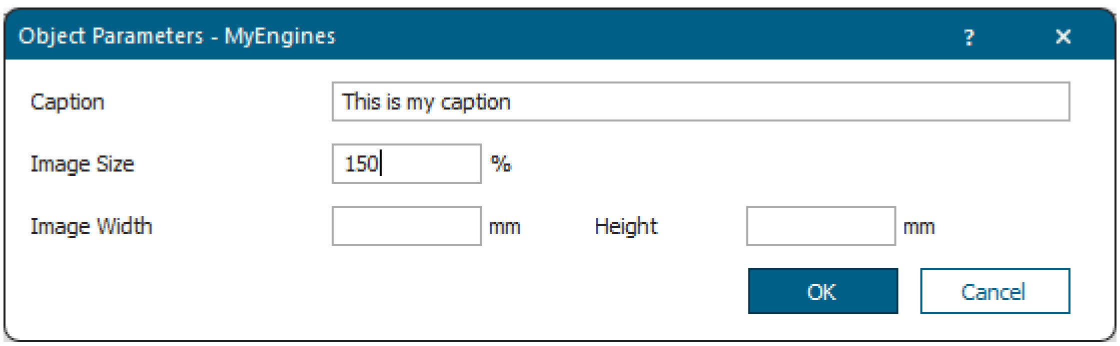
Type Object Parameters in the Dialog Object Parameters

To display a screenshot of a **Frame**, which contains the icons of all objects inserted into this *Frame*, on the HTML page, click  on the toolbar .

Then select the *Frame* whose screenshot you would like to show and click **OK**.

Type in the **Caption** which the *HtmlReport* shows below the *Frame*. If you do not type in a **Caption**, the report shows the path to the *Frame* as the **caption**.

Type in the relative **Image Size** in percent to which the *HtmlReport* scales the screenshot of the object. Instead, you can also type in the **Image Width** and the **Image Height**. You can also type in an asterisk * for either the **Width** or the **Height** to ensure that the image will not be distorted. The relative size takes precedence over the absolute size.



Type the Parameters on the Tab Content

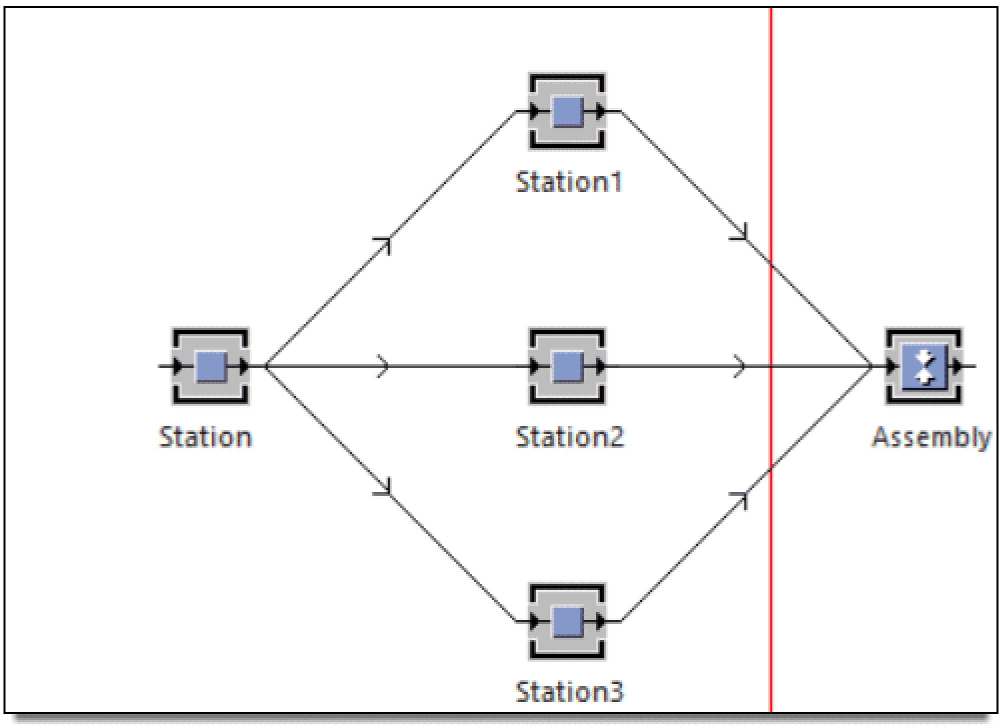
You can also set the size of the screenshot on the HTML page by typing in these instructions:

- Type in an asterisk * to zoom the screenshot out, when you zoom the report window out. When zooming in, the icon is never enlarged to more than its original size.
- Type in a number and the percent sign % to scale the screenshot with this value.
- Type in the width and the height of the image in millimeters to show the screenshot with these measurements. You can also type in an asterisk* for either the **Width** or the **Height** to ensure that the image will not be distorted.

```
[MyEngines] // our Frame, which we inserted into ‘MyPlant’, is called
‘MyEngines’
[MyEngines, *]
[MyEngines, 150%]
```

```
[MyEngines, 100, 150]
[MyEngines, 100, *]
[MyEngines, *, 150]
[MyEngines, "This is my caption"]
[MyEngines, "This is my caption", *]
[MyEngines, "This is my caption", 150%]
[MyEngines, "This is my caption", 100, 150]
[MyEngines, "This is my caption", 100, *]
[MyEngines, "This is my caption", *, 150]
```

The result looks like this:



This is my caption

If you create a *user-defined attribute* named *ReportObject* for an object, *Plant Simulation* does not display the object itself anymore, but its *ReportObject* instead. If the *user-defined attribute* has the data type *object*, *Plant Simulation* shows the object which the attribute references. In many cases this will be a *HtmlReport* which inserts its report into the HTML page. To insert a *DataTable* or the result of a method call, you do not have to reference an object of type *DataTable* or *Method*, but you can set the data type of the *user-defined attribute* to *table* or *method* respectively and define the *DataTable* or *Method* respectively in the attribute.

If you want to reference another object in a *Frame*, you do not have to create a *user-defined attribute* of data type *object*, but you can name an inserted object `ReportObject`. This object will then be shown instead of the *Frame* on the HTML page.

In this case only the parameters of the *HtmlReport* apply, not those of the *2D Frame*, or of the *3D Frame*.

Specify Named Arguments

The *HtmlReport* provides the same named arguments as a **picture of a 3D scene** for showing a *Frame*.

See also

Show Statistics and Other Values in a Report

Display a HtmlReport

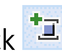
getHTMLCode [SimTalk] - Frame

getHTMLCode [SimTalk] - Chart

Display a GanttChart

Specify the settings for displaying an object of type *GanttChart* on the HTML page.

Select the Object in the Dialog Select Object

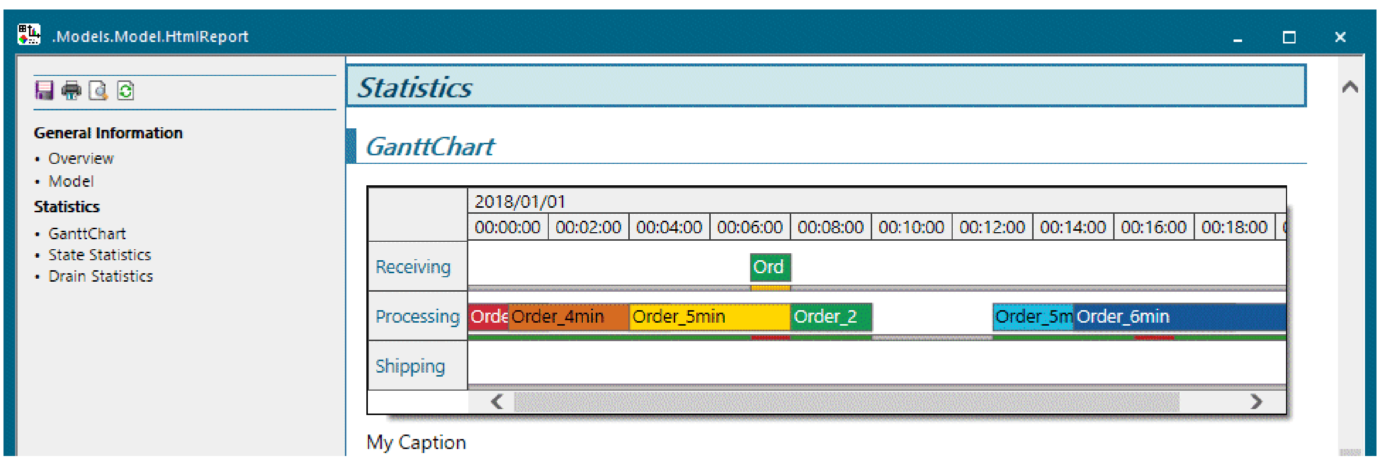
To display the diagram picture of a *GanttChart* on the HTML page, click  on the toolbar . Then select the *GanttChart* in the dialog **Select Object** and click **OK**.

Specify the Parameters on the Tab Content

Instead, you can also type in the instructions that correspond to the settings you can enter into the dialog.

```
## GanttChart
[GanttChart, "My Caption", 0.5]
// adds our GanttChart with a time scale of 0.5, with the caption My
Caption.
```

The result looks like this:



Drag the scroll bar to the right to scroll to the right in the *GanttChart*.

Specify Named Arguments

You can specify the following named arguments:

- The named argument *Caption* specifies the caption with which the *GanttChart* is shown in the *HtmlReport*.

```
[GanttChart Caption="My Caption"]
```

- The named argument *Scale* sets the scaling on the time axis with which the *GanttChart* is shown in the *HtmlReport*.

```
[GanttChart Scale=0.5]
```

See also

GanttChart [object]

Show Statistics and Other Values in a Report

Show Reports in a Hierarchically Structured Model

exportChart [SimTalk]

Display a HtmlReport

Specify the settings for displaying an object of type *HtmlReport* on the HTML page.

Select the Object in the Dialog Select Object

To display an *HtmlReport* on the HTML page, click  on the toolbar  and select it or type its name within brackets. Then select the *HtmlReport* which you would like to show and click **OK**.

Specify the Parameters on the Tab Content

You can specify an object as argument for the *HtmlReport*. You can then access the object in the called *HtmlReport* by typing in **@** within brackets, `[@]`. This is especially useful if you want to display several objects in the same way.

At times, you might also display an *HtmlReport* on its own and also embed it as part of a larger report. Suppose that you model a production line in a *sub-Frame* and assign this *sub-Frame* an *HtmlReport*. In this case you might want to display the *HtmlReport* individually on its own but also integrate it into the *HtmlReport* of the surrounding *Frame*. Then you might want to treat the heading for the sub-report different from for the individual report. You might, for example, want to display headings of level 1 in the sub-report as headings of level 2 and headings of level 2 as headings of level 3, etc. Then you can specify the numbers sign/hash # in the *HtmlReport*. You can also specify several number signs ##. The *HtmlReport* increases the heading levels by 1 per a single number sign, meaning that the size of the heading levels is decreased.

```
[HtmlReport2]
[HtmlReport2, Station1]
[HtmlReport2, Station1, #]
```

If you create a *user-defined attribute* named *ReportObject* for an object, *Plant Simulation* does not display the object itself anymore, but its *ReportObject* instead. If the *user-defined attribute* has the data type *object*, *Plant Simulation* shows the object which the attribute references. In many cases this will be a *HtmlReport* which inserts its report into the HTML page. To insert a *DataTable* or the result of a method call, you do not have to reference an object of type *DataTable* or *Method*, but you can set the data type of the *user-defined attribute* to *table* or *method* respectively and define the *DataTable* or *Method* respectively in the attribute.

If you want to reference another object in a *Frame*, you do not have to create a *user-defined attribute* of data type *object*, but you can name an inserted object `ReportObject`. This object will then be shown instead of the *Frame* on the HTML page.

You can also create sub-reports for all instances of an object. To do so, type in the path to the object, followed by an asterisk *) as parameter. *Plant Simulation* then calls the specified *HtmlReport* for all instances which inherit directly or indirectly from the object, and which are inserted into the *Frame*, in which the outer *HtmlReport* is located. This also includes *sub-Frames*.

```
[HtmlReport2, .MaterialFlow.Station*]
```

See also

HtmlReport [object]

Show Statistics and Other Values in a Report

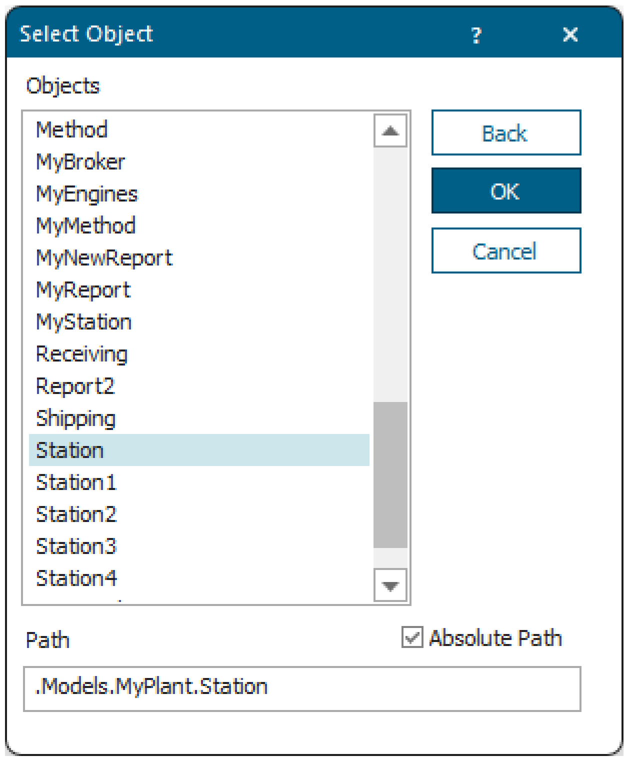
Collect Object Instances and Show Them

Display an Icon of an Object

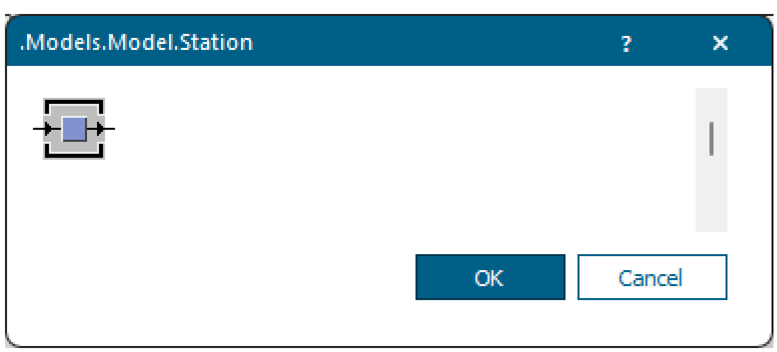
Specify the settings for displaying an *icon of an object* on the HTML page.

Type Object Parameters into the Dialog Object Parameters

To display the **icon of an object** on the HTML page, click  on the toolbar . Then select the object whose icon you want to show in the dialog **Select Object** and click **OK**.



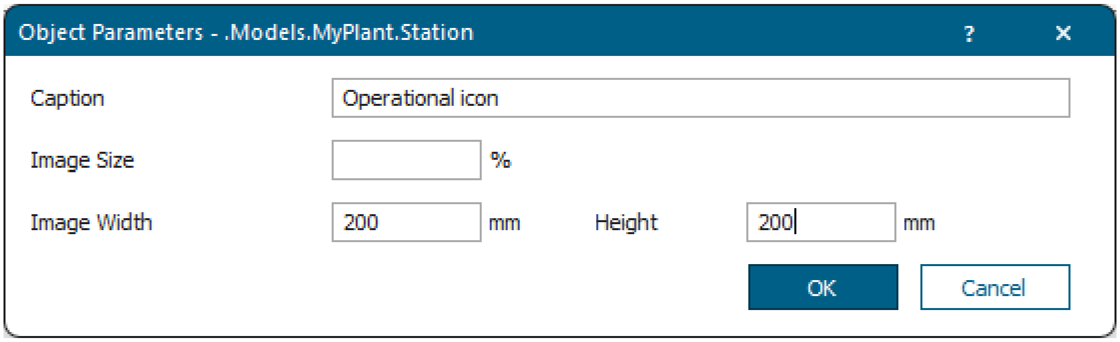
Select the icon you want to show and click **OK**.



Type in the **Caption** which the *HtmlReport* shows below the icon.

Either type in the **Image Size** in percent.

Or type in the **Image Width** and the **Image Height**.



The settings in the dialog **Object Parameters** translate to this statement:

```
[!.Models.Model.Station, "Operational", "Operational icon", 200, 200]
```

Specify Named Arguments

> **Note**
>
> When specifying named arguments to display an object icon, you have to type in an exclamation mark `!` in front of the actual statement.

You can specify the following named arguments:

- The named argument *Icon* specifies the icon of the object that is going to be shown in the *HtmlReport*.

  Specify a *string* to show the icon with the specified name of the specified object.

  Specify a *number* to show the icon with the specified number of the specified object. The number 2 for example shows the second icon of the object.

  If you not specify the named argument *Icon*, *Plant Simulation* shows the active icon.

- The named argument *Caption* specifies the caption with which the *specified icon of the object* is shown in the *HtmlReport*.

- The named argument *Size* specifies the size of the icon.

  You can specify the size like this:

  - Specify `*` or `[*, *]` or do not specify the argument to show the icon with its default size.
  - Specify `[*, y]` to set the height of the icon to y and to use the width of the default width of the icon.
  - Specify `[x, *]` to set the width of the icon to x and to use the height of the default height of the icon.

- Specify a percentage to scale the icon by the specified factor.
- Specify `[x, y]` to scale the icon to the size x and the height y. This might distort the icon though.

```
[!Station Icon=Operational]
```

Display a Method

Specify the settings for displaying an object of type *Method* on the HTML page.

Specify the Parameters on the Tab Content

To run the **Method**, whose name you type in brackets, and to display the return value on the HTML page, type in the following instructions.

Within parentheses, you can also pass parameters to the *Method*.

```
[Method]
[Method(Argument1, Argument2])
```

The source code of our *Method* named *MyMethod* queries the **x-position** and the **y-position** of the object named *MyStation*.

The source code looks like this:

```
param obj: object -> string
return to_str("x=", obj.XPos, ", y=", obj.YPos)
```

The following statement displays these positions in the *HtmlReport*.

```
[MyMethod(MyStation)]
```

The result looks like this:

x=103, y=160

See also

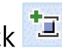
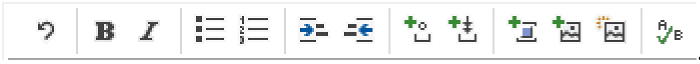
Method

Show Statistics and Other Values in a Report

Display a PythonModule

You might want to display an object of type *PythonModule* on the HTML page in the *HtmlReport* to make use of the vast array of charting and display libraries that are available for *Python*.

Select the Object in the Dialog Select Object

To display a *PythonModule*, which you inserted into your simulation model, on the HTML page, click  on the toolbar . Then select the *PythonModule* in the dialog **Select Object** and click **OK**.

The *HtmlReport* then adds `[PythonModule]` to the **Tab Content**.

If you do not specify a function, the *PythonModule* shows its source code in the *HtmlReport*.

Specify Named Arguments

The named argument *Call* specifies the function that is going to be called and whose return value, provided it has the data type *string*, is to be shown in the *HtmlReport*.

```
[PythonModule Call="multiply"]
```

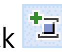
See also

PythonModule

Display a SankeyDiagram

Specify the settings for displaying an object of type *SankeyDiagram* on the HTML page.

Select the Object in the Dialog Select Object

To display the diagram picture of a *SankeyDiagram* on the HTML page, click  on the toolbar . Then select the *SankeyDiagram* in the dialog **Select Object** and click **OK**.

Type in the Parameters on the Tab Content

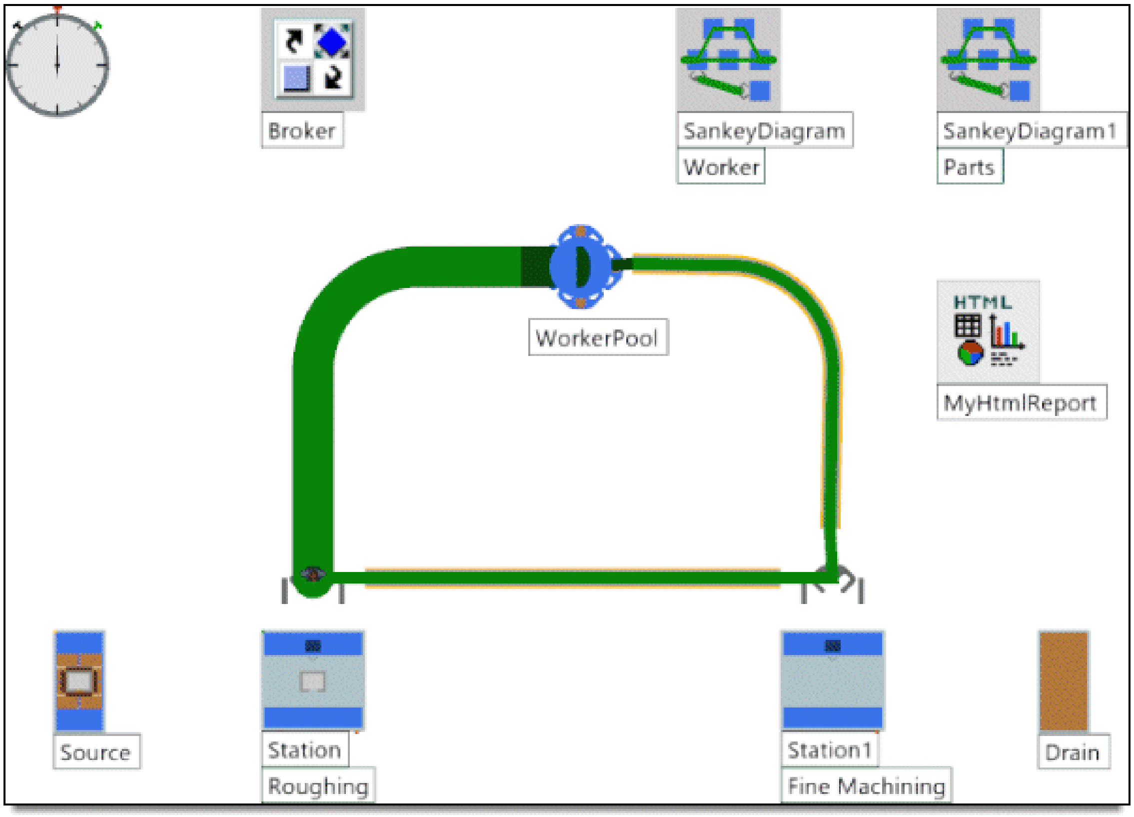
Instead, you can also or type in the following instructions. The syntax resembles the syntax of the *Frame*, as the *SankeyDiagram* also depicts *Frame* images.

```
Path[, Title][, SizeParameters]
```

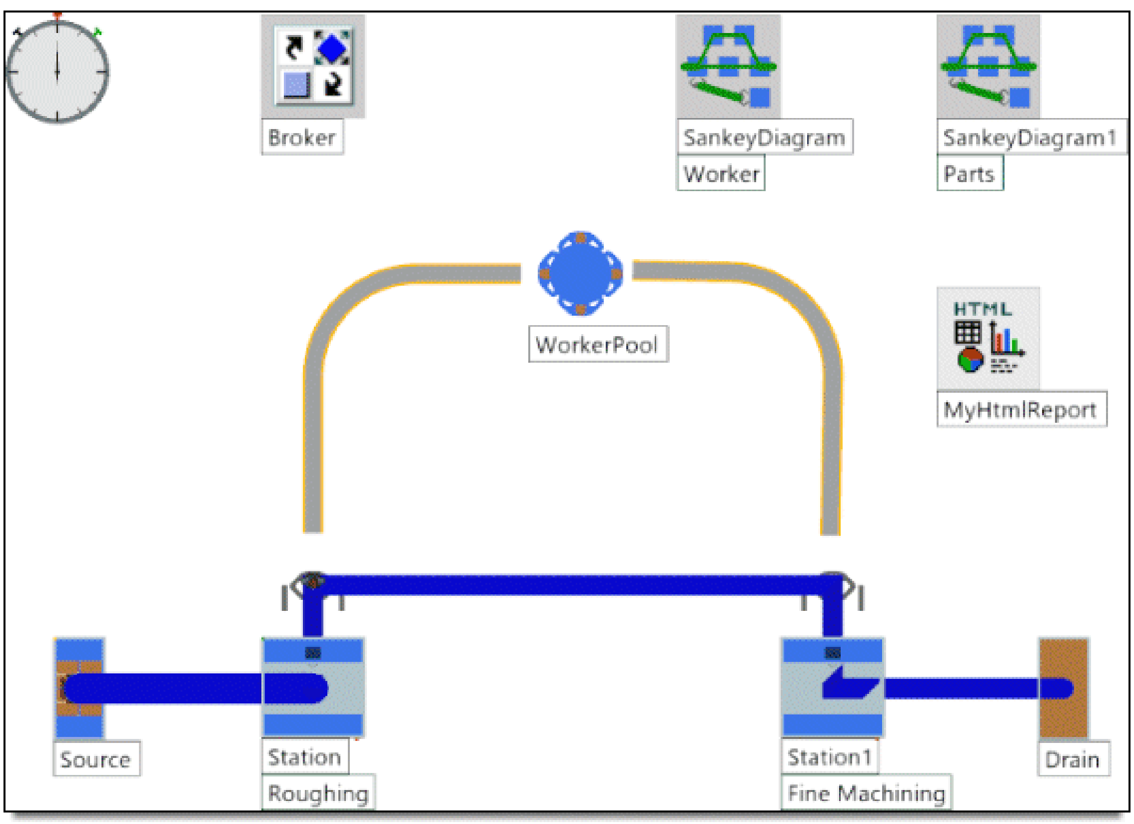
You might, for example, type in these instructions:

```
[SankeyDiagram, Workers Sankey flows, 90, 90]
[SankeyDiagramm1, Parts Sankey flows, 85%]
```

If we do not specify a size, the depiction looks best though:



Worker Sankey flows



Parts Sankey flows

You can type in the camera position and the camera rotation in addition:

```
Path[, Title][, SizeParameters][[CameraPosX, CameraPosY, CameraPosZ],
CameraRotX, CameraRotZ]
```

If you do not specify any camera parameters, *Plant Simulation* shows the *SankeyDiagram* in the **Planning View**.

Specify Named Arguments

A *SankeyDiagram* expects the same named arguments as a **picture of a 3D scene**. If you do not specify the argument **PlanningView**, a *SankeyDiagram* will be shown in the **PlanningView**.

- The named argument `Type "exclusive"` specifies that the *SankeyDiagram* is shown by itself in the *HtmlReport*.

  ```
  [SankeyDiagram Type="exclusive"]
  ```

- The named argument `Type "additional"` specifies that the *SankeyDiagram* is shown together with the other visible *Sankey Diagrams* in the *HtmlReport*.

  ```
  [SankeyDiagram Type="additional"]
  ```

- The named argument `Type "all"` specifies that all *Sankey Diagrams* of the *Frame*, in which the *SankeyDiagram* is located, are shown in the *HtmlReport*.

  ```
  [SankeyDiagram Type="all"]
  ```

  Default Value of the Parameter

  The default value is "exclusive".

Note

The named arguments *Type "exclusive", Type "additional", Type "all"* only work in simulation models with the model language **English, Chinese, Japanese**, and **Hungarian**.

The named arguments *Type "exklusiv", Type "zusätzlich", Type "alle"* only work in simulation models with the model language **German**.

See also

SankeyDiagram

Display a Frame

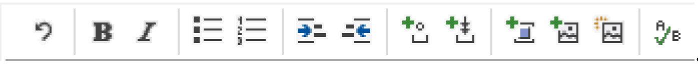
Show Statistics and Other Values in a Report

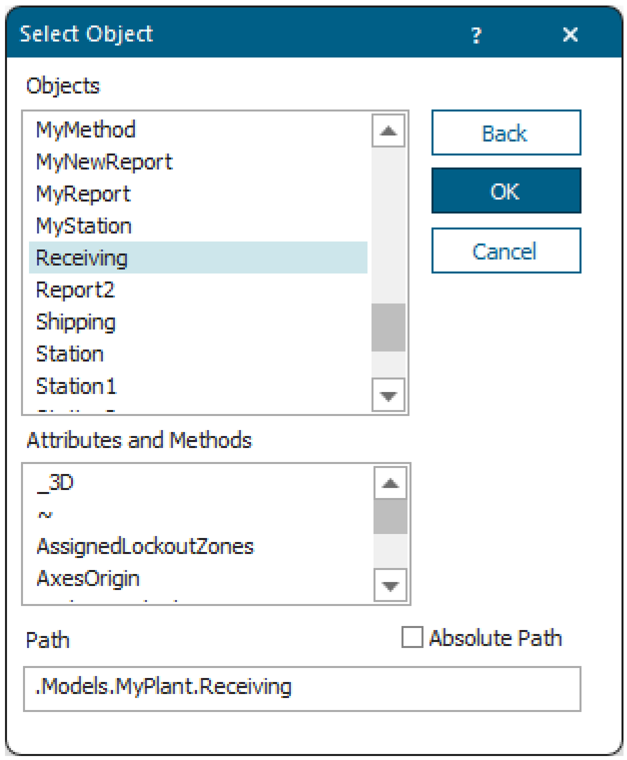
Show Reports in a Hierarchically Structured Model

Display Statistics Values of an Object as a Table

Specify the settings for displaying statistics values of an object as a table on the HTML page.

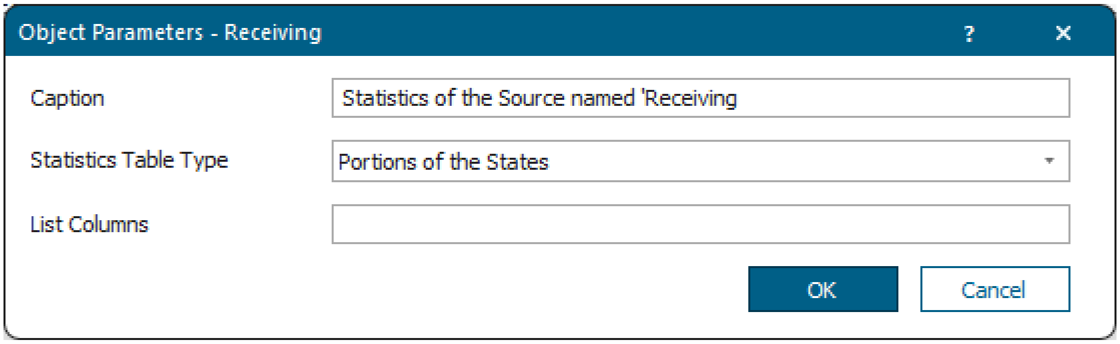
Select the Object in the Dialog Select Object

To display statistics values as a table, click  on the toolbar . Then select the object for which you would like to show statistics values. Click **OK**.



Type in the **Caption** which the *HtmlReport* shows below the statistics table on the HTML page.

Select the **Statistics Table Type** you want to show for the object. The different object types provide different table types. If you do not type anything into the text box **List Columns**, the *HtmlReport* shows all portions.



The settings above result in this source code:

```
[Receiving, "Statistics of the Source 'Receiving'", %Stati]
```

The result might look like this:

Object	Working	Set-up	Waiting	Blocked	Powering up/down	Failed	Stopped	Paused	Unplanned	Portion
Receiving	0.00%	0.00%	0.00%	94.79%	0.00%	5.21%	0.00%	0.00%	0.00%	

State statics of the Source 'Receiving'

To only show certain columns, type in their column index, separated by commas, into the text box **List columns**, for example.

The settings above result in this source code:

```
[Receiving, "Statistics of the Source 'Receiving", %Stati, 4,6]
```

The result might look like this:

Object	Blocked	Failed
Receiving	94.79%	5.21%

State statics of the Source 'Receiving'

> **Note**
>
> If the selected object collected statistics values, the *HtmlReport* shows values in a table. If the object did not collect statistics, the *HtmlReport* naturally cannot show any data.

If the *HtmlReport* lists several objects one below the other, it aggregates their values if possible, i.e., it lists their values row-by-row in a table. If the *HtmlReport* cannot aggregate the values, it shows the non-aggregated values and starts a new table. If you want to show instances of a class object, the *HtmlReport* lists the values of all instances row-by-row.

As statistics data can cover various values, the objects provide a number of different settings you can select in the drop-down list **Statistics Table Type**:

Drop-down list item	corresponds to this text you can type in, English	corresponds to this text you can type in, German
Portions of the States	%ResStates	%Stati
Material Flow Properties	%MatFlowProperties	%Matflusseigenschaften
Rotation Time	%RotationTime (*Turnplate*, *Turntable*, *PickAndPlace*)	%Drehungszeit (*Drehplatte*, *Drehtisch*, *PickAndPlace*)
Moving Time	%MovingTime (*Converter*)	%Umsetzzeit (*Umsetzer*)
Working Time	%WorkingTime	%Arbeitszeit
Set-up Time	%SetupTime	%Rüstzeit
Waiting Time	%WaitingTime	%Wartezeit
Blocked Time	%BlockedTime	%Blockiertzeit
Powering up/down Time	%PowerUpDownTime	%HochHerunterfahrzeit
Stopped Time	%StoppedTime	%Angehaltenzeit
Failed Time	%FailedTime	%Störungszeit
Paused Time	%PausedTime	%Pausenzeit
Empty Time	%EmptyTime	%Leerzeit
Material Flow Properties of Mobile Units	%MUMatFlowProperties	%BEMatflusseigenschaften
Empty Time of Mobile Units	%MUEmptyTime	%BELeerzeit
Cumulated Statistics of the Parts Which the Drain Deleted	%DrainCumulated	%SenkeKumuliert
Part Types Which the Drains Deleted	%DrainAllTypes	%SenkeAlleTypen
Detailed Portions of the Part Types which the Drain Deleted	%DrainTypesPortions (non summable)	%SenkeTypenanteile (nicht addierbar)
Detailed Times of the Part Types which the Drain Deleted	%DrainTypesTimes (non summable)	%SenkeTypenzeiten (nicht addierbar)
Total Consumption and Portions of the Energy States	%Energy	%Energie
Product-Oriented Statistics of all Existing and Deleted MUs (by Classes)	%MUClassStatistics	%BEKlassenStatistik
Mean Time Portions of a MU Class of the Mean Life Span	%MUClassTimePortions	%BEKlassenZeitanteile

Drop-down list item	corresponds to this text you can type in, English	corresponds to this text you can type in, German
Mean Time Portions of a Individual MU of the Mean Life Span	%MUTimePortions	%BEZeitanteile
Usage of the Transporters	%TransUsageTime	%TransVerwendungszeit
Ready Times	%TransReadyTime	%TransBereitzeit
Pauses	%TransPausedTime	%TransPausenzeit
Failures	%TransFailedTime	%TransStörungszeit
Traveled Distance	%TransTraveledDistance	%TransWegstrecke
Battery	%TransBatteryTime	%TransBatteriezeit
Broker Statistics	%Broker	%Broker
Mediation Time	%BrokerMediationTime	%BrokerVermittlungsdauer
Dwelling Time	%BrokerDwellTime	%BrokerVerweildauer
Capacities and States of the Exporters	%Exporter	%Exporter
Time Portions of the Exporters	%ExporterTimePortions	%ExporterZeitanteile
Traveled Distance by Workers	%WorkerTraveleledDistance	%WerkerWegstrecke
Importers Waiting for Services and Parts	%ImporterWaiting	%ImporterWartend
Waiting Times for Services and Parts	%ImporterWaitingTime	%ImporterWartezeit
Waiting Times for Parts	%ImporterMUWaitingTime	%ImporterBEWartezeit
Waiting Times for Set-up Exporters	%ImporterSetpWaitingTime	%ImporterRüstWartezeit

If you did not select a **Statistics Type Table**, the *HtmlReport* shows these default statistics types:

Object	Default Type, English	Default Type, German
Part	%MUTimePortions	%BEZeitanteile
Container	%MUMatFlowProperties	%BEMatFlusseigenschaften
Transporter	%TransUsageTime	%TransVerwendungszeit
Worker	%WorkerTraveledDistance	%WerkerWegstrecke
Turnplate, *Turntable*, *PickAndPlace*	%RotationTime	%Drehungszeit
Converter	%MovingTime	%Umsetzzeit
Drain	%DrainCumulated	%SenkeKumuliert

Object	Default Type, English	Default Type, German
Exporter	%Exporter	%Exporter
Broker	%BrokerMediationTime	%BrokerVermittlungsdauer
All other objects	%States	%Stati

Specify the Parameters on the Tab Content

Instead of selecting the **Statistics Table Type** in the dialog and typing the caption, you can also type the instructions into the tab **Contents** of the dialog of the *HtmlReport*:

```
[Object path, caption, statistics table type]
```

You might, for example, type in:

```
[Receiving, "Statistics of the Source 'Receiving'", %States]
[Station, "My values", %States, 1..4]                  // shows columns 1 to
4
[Station, "My values", %States, #Working, #SettingUp] // shows columns 1
and 2
```

See also

Display a Drain

Display a Broker

Show Statistics and Other Values in a Report

Display a Table from the SQLite Interface

Specify the settings for displaying a table containing data of a table of the *SQLite* interface on the HTML page.

Select the Object in the Dialog Select Object

To display data of a table of the **SQLite** interface on the HTML page, click  on the toolbar . Then select the *SQLite* interface in the dialog **Select Object** and click **OK**.

Specify the Parameters on the Tab Content

This instruction displays the contents of the table called *MUTrace*.

```
[MySQLite, "select * from MUTrace"]
```

Specify Named Arguments

You can specify the following named arguments:

- The named argument *Statement* specifies the SQL statement of the *SQLite* object which is shown in the *HtmlReport*.

  ```
  [MySQLite Statement="select * from MUTrace"]
  ```

- The named argument *MaxRows* shows the maximum number of rows which the *SQLite* object shows in the *HtmlReport*.

  ```
  [MySQLite MaxRows=5]
  ```

See also

SQLite

Show Statistics and Other Values in a Report

Display a Variable

Specify the settings for displaying the contents of a *Variable*, according to the type of its contents on the HTML page.

Select the Object in the Dialog Select Object

To display the contents of a *Variable* on the HTML page, click  on the toolbar . Then select the *Variable* in the dialog **Select Object** and click **OK**.

If the *Variable* has the data type **real**, **length**, **money**, **weight**, **time**, **speed**, or **acceleration**, you can specify the **Number of Decimal Places** and/or the **Number of Integer Places** to which the value will be rounded as an optional parameter.

If the *Variable* contains a *DataTable/List object*, it will be treated like a reference to a *DataTable/List object*, and you can parametrize it just like these. In the other case you cannot enter any parameters.

If the *Variable* references an object, *Plant Simulation* displays the path of the object. If this object does not exist, the report shows a question mark (**?**) after the **Path**.

Plant Simulation inserts all other values, in some cases with the respective unit, as text.

```
[Variable]
[Variable, "Tab. 1: List of all tabular values"]
[Variable, 0, 2..4, 7]
[Variable, *..4, 7]
[Variable, 0, 2..4, 7..*]
[>Variable] // Shows the object specific content instead of the
            // object path if the variable references an object.
```

Specify Named Arguments

The named argument *Precision* specifies the **Number of Decimal Places** and/or the **Number of Integer Places** with which the *Variable* is shown in the *HtmlReport*.

```
[Variable Precision=5]
```

See also

Variable

Show Statistics and Other Values in a Report

Display a Picture of a Scene

Specify the settings for displaying a picture of a scene on the HTML page.

Type in the Parameters on the Tab Content

```
[Object._3D]
[Object._3D, w, h]
[Object._3D, [caption,] w, h]
[Object._3D, w, h, crx, crz]
[Object._3D, [caption,] w, h, crx, crz]
[Object._3D, w, h, cpx, cpy, cpz, crx, crz]
[Object._3D, [caption,] w, h, cpx, cpy, cpz, crx, crz]
```

Note

You can also embed `_3D` without specifying any additional arguments at all.

To display a picture of a 3D scene on the HTML page, type in the following statement:

```
[Object._3D, w, h]
[Object._3D, [caption,] w, h]
[Object._3D, w, h, crx, crz]
[Object._3D, [caption,] w, h, crx, crz]
[Object._3D, w, h, cpx, cpy, cpz, crx, crz]
[Object._3D, [caption,] w, h, cpx, cpy, cpz, crx, crz]
```

The setting `[Object._3D, w, h]` creates the picture with the angles crx=45° and crz=15° and as a **View All** view.

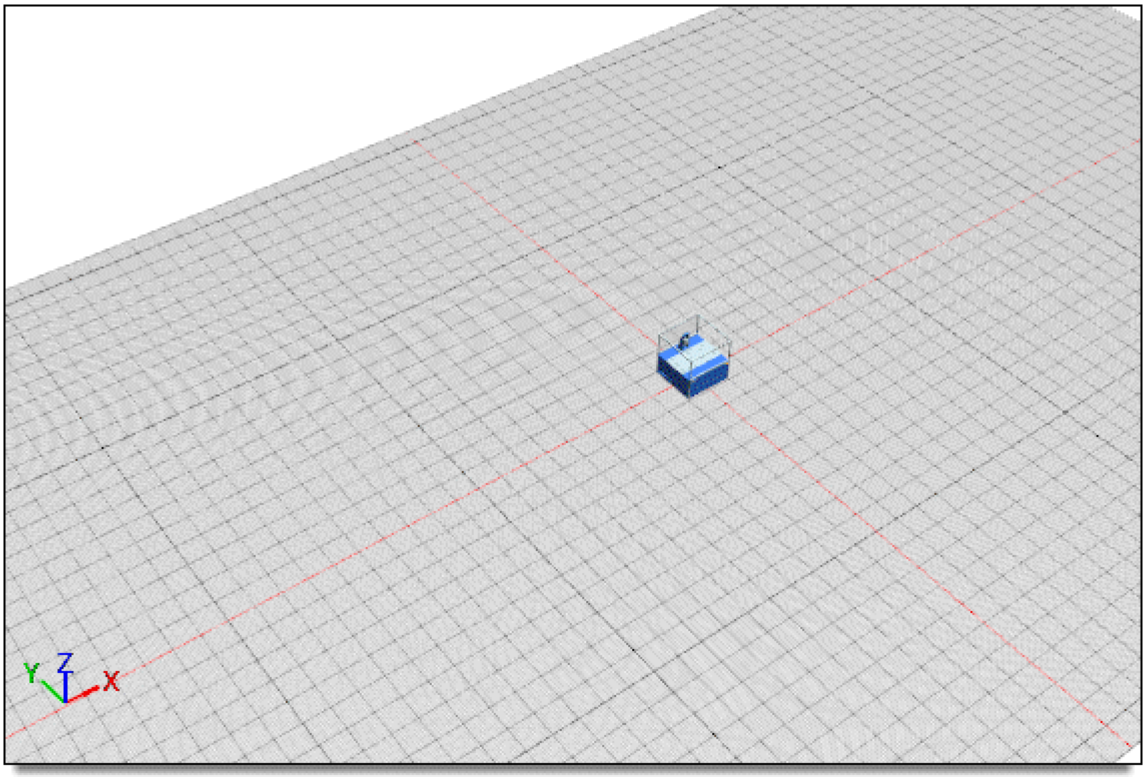
The individual arguments designate the following settings:

- The optional argument `caption` designates the caption which the *HtmlReport* shows below the picture.
- The named argument `object` designates the object whose picture you want do display.
- The named argument `w` designates the width of the picture in pixels.
- The named argument `h` designates the height of the picture in pixels.
- The named argument `cpx` designates the x-coordinate of the camera position.
- The named argument `cpy` designates the y-coordinate of the camera position.
- The named argument `cpz` designates the z-coordinate of the camera position.
- The named argument `crx` designates the rotation angle of the camera on the x-axis.
- The named argument `crz` designates the rotation angle of the camera on the z-axis.

You might, for example, type in:

```
[Station._3D, 150, 100, -29.596, -31.985, 39.985, 46.983, -36.669]
```

The result looks like this:

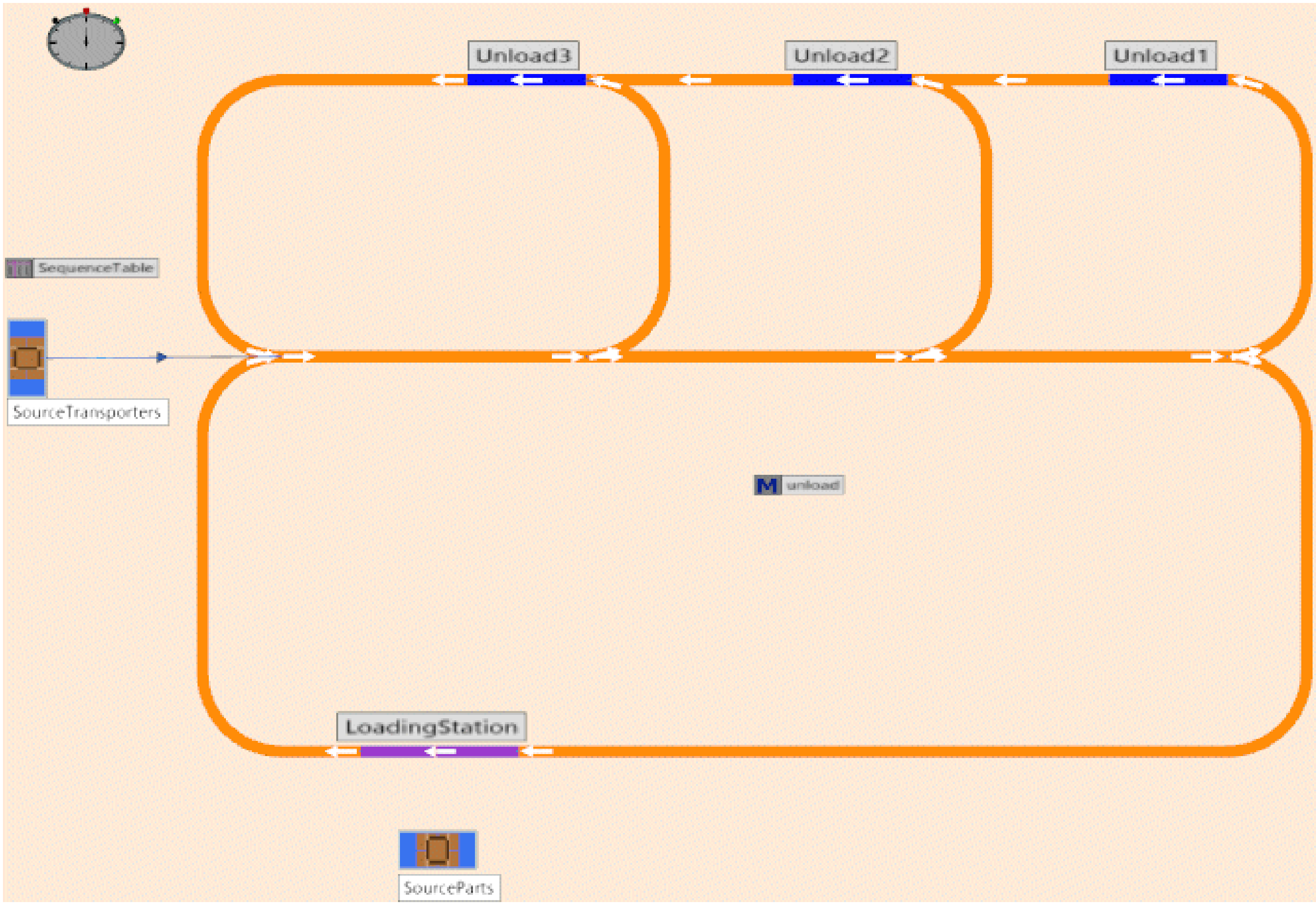


.Models.MyPlant.Station

The coordinates are those that the command **Marks** > **Generate SimTalk Code and Copy to Clipboard** on the **View** ribbon tab of the *3D window* copies.

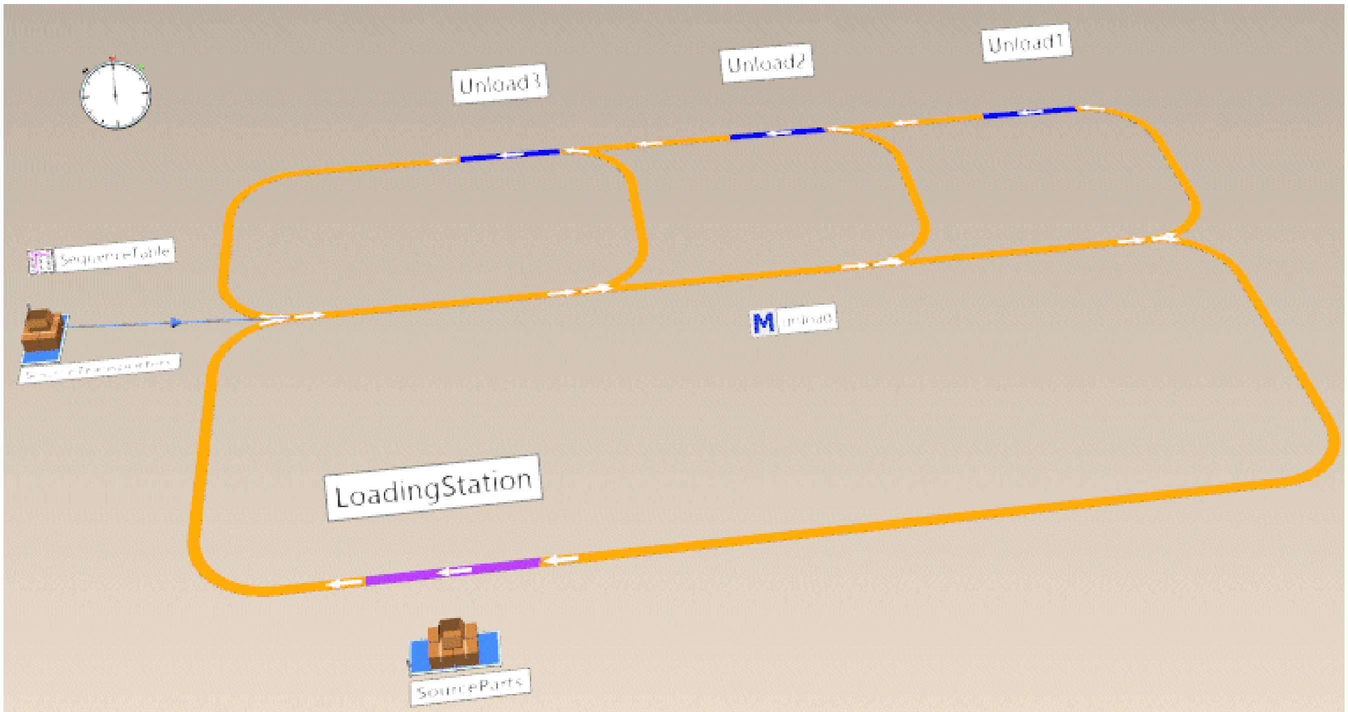
For a scene you can select the command **Generate HtmlReport Code and Copy to Clipboard** and then paste the code into the *HtmlReport*.

- If **Planning View** is active, *Plant Simulation* generates the respective code.




```
[.Models.Model._3D Size=[874,491] PlanningView=true
ViewPos=[21.483,-11.597] ViewZoom=23.294]
```

- If **Planning View** is not active, *Plant Simulation* uses the current camera position or camera rotation in the 3D scene respectively and copies it to the generated code.




```
[.Models.Model._3D Size=[874,491] PlanningView=false
CameraPos=[15.211,-46.816,29.319] CameraRotX=48.000 CameraRotZ=-8.500]
```

You can **access the 3D window**.

To allow this, we support the parameters for the rotation or for the rotation and position of the camera in a *Frame* in the simulation model:

```
[Frame[,Caption][, {Width, Height} {Percent}][, CameraPositionX,
CameraPositionY, CameraPositionZ, CameraRotationX, CameraRotationZ}
{CameraRotationX, CameraRotationZ}]]
```

If you specify the camera rotation, you have to type in two asterisks instead of one asterisk:

```
[current, *, *, -45.0]
```

If you specify the camera position and the camera rotation, you have to type in two asterisks instead of one asterisk:

```
[current, *, *, -7.12, 8.123, 1.24, -45.0, 15,0]
```

Instead of the camera information you can also specify the keyword `%planningview` in **3D scenes**. Then the *HtmlReport* shows the **3D scene** in the **planning view**.

```
[current, *, *, %planningview]
```

You can show the views of 3D scenes in the *HtmlReport* with the keyword **%modelingview**.

If a **3D scene** is shown in **Planning View**, you can circumvent this state and show the modeling view instead of the planning view with the keyword **%modelingview** in the *HtmlReport*.

```
[current, *, *, %modelingview]
```

Specifying Named Arguments

The *HtmlReport* provides these arguments for showing a picture of a 3D scene:

- The named argument *Caption* specifies the caption with which the picture is shown in the *HtmlReport*.

  ```
  [.Models.Model._3D Caption="My Caption" Size=[874,491] PlanningView=true]
  ```

  Default Value of the Argument

  The default value is the object path.

- The named argument *PlanningView* specifies if the picture to be shown in the *HtmlReport* is created in **Planning View** (`true`) or in **Modeling View** (`false`).

  ```
  [.Models.Model._3D Caption="My Caption" Size=[874,491] PlanningView=true]
  ```

  Default Value of the Argument

  The default value is the setting you selected for the **View Options** of the *Frame*. Otherwise, it is false.

- The named argument *Size* sets the size with which the picture is shown in the *HtmlReport*.

  Specify * (asterisk) or `[*, *]` to make *Plant Simulation* set the size automatically.

  Specify `[*, Height]` or `[Width, *]` to make *Plant Simulation* set the size designated by * via the default aspect ratio from the other size.

  Specify `[Width, Height]` to make *Plant Simulation* set the width and the height of the picture.

  The default value is 700 by 500 pixels.

  ```
  [.Models.Model._3D Caption="My Caption" Size=[874,491] PlanningView=true]
  ```

- The named argument *ScaleTo* specifies the scaling of the picture relative to the width of the *HtmlReport* in percent.

  Default Value of the Argument

  The default value is 100%.

  ```
  [.Models.Model._3D Caption="My Caption" ScaleTo=80% PlanningView=true]
  ```

The *HtmlReport* provides these additional arguments for showing a picture of the **planning view** of a 3D scene:

- The named argument *ViewPos* sets that the picture is shown centered in the *HtmlReport*.
- The named argument *ViewZoom* sets the zoom factor or the view that is shown in the *HtmlReport*.

```
[.Models.Model._3D Size=[874,491] PlanningView=true
ViewPos=[21.483,-11.597] ViewZoom=23.294]
```

The *HtmlReport* provides these additional arguments for showing a picture of the **modeling view** of a 3D scene:

- The named argument *CameraPos* sets the camera position with which the picture is shown it in the *HtmlReport*.
- The named argument *CameraRotX* sets the camera rotation in degrees around the x-axis with which the picture is shown in the *HtmlReport*.
- The named argument *CameraRotZ* sets the camera rotation in degrees around the z-axis with which the picture is shown in the *HtmlReport*.

```
[.Models.Model._3D Size=[874,491] PlanningView=false
CameraPos=[15.211,-46.816,29.319] CameraRotX=48.000 CameraRotZ=-8.500]
```

See also

Generate HtmlReport Code and Copy to Clipboard

Generate SimTalk Code and Copy to Clipboard [camera mark]

Compute a Formula

To compute any formula, type in the formula between an opening bracket, followed by the equal sign `[=` and a closing bracket `]`.

Remarks

You can type in the number of decimal places, separated by a comma, as parameter. To force a certain number of decimal places, even when these are zeros, type in a space followed by zero `0` in front of the number of decimal places.

```
[=Station1.statWorkingPortion + Station1.statBlockingPortion]
[=Station1.statWorkingPortion + Station1.statBlockingPortion, 2]
[=Station1.statWorkingPortion + Station1.statBlockingPortion, 02]
```

The result looks like this:

```
1
1
1.00
```

If you query an attribute or compute a formula, and if the return value has the data type *object*, *Plant Simulation* displays the object path on the HTML page. If you want to display the object itself instead of the object path on the HTML page, type in the attribute or the formula between `[>` and `]`.

```
[>Source.path]  // displays the table of the delivery list
[>@.~]  // displays the location of @m (for example the contents
        // of the Frame when @ is a Station inserted into the Frame)
[>=str_to_obj(Variable)]
```

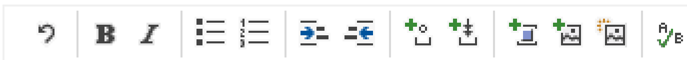
See also

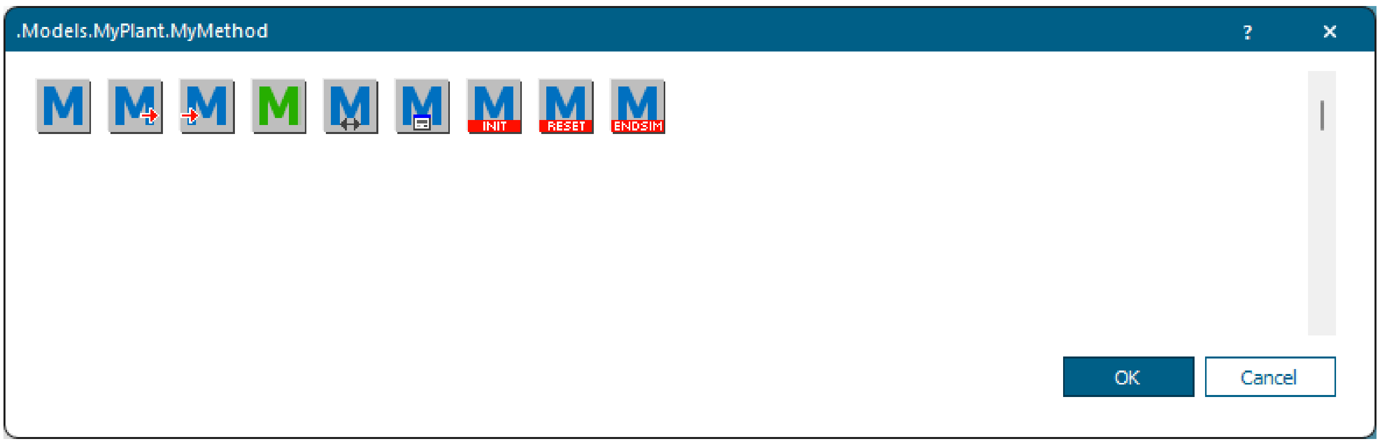
Show Statistics and Other Values in a Report

Add an Image of an Object

Specify the settings for displaying the current icon of an object, or for displaying an object and an icon on the HTML page.

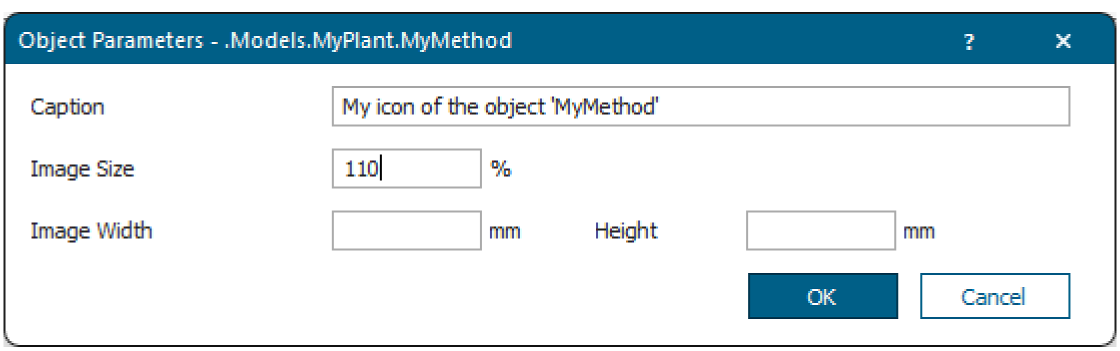
Select the Object in the Dialog Select Object

To display an object to display its current icon, or an object and an icon to be displayed on the HTML page, click  on the toolbar . Select the icon you want to use in the dialog that opens and click **OK**.



Type in the **Caption** which the *HtmlReport* shows below the picture.

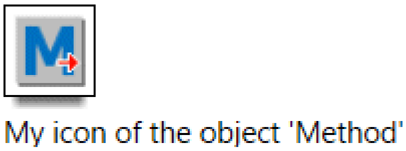
Type in the **Image Size** in percent to which the *HtmlReport* scales the picture. Instead, you can also type in the **Image Width** and the **Image Height**. You can also type in an asterisk (*) for either the **Width** or the **Height** to ensure that the image will not be distorted. The relative size takes precedence over the absolute size.



The settings above result in this statement:

```
[!MyMethod, "ExitCtrl", "My icon of the object 'Method'", 110%]
```

The result looks like this:



If the text cursor is located within a picture reference ([!...]), *Plant Simulation* first opens the dialog **Open File** and then the dialog **Object Parameters** for the path within brackets when you click **Add Icon** .

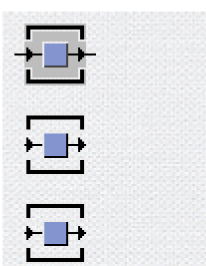
If the text cursor is located within a picture reference ([!...]), *Plant Simulation* first opens the dialog **Open File** and then the dialog **Object Parameters** in which the path is preselected that was detected within brackets when you hold down **Shift** while clicking **Add Icon** .

Specify the Parameters on the Tab Content

Instead, you can also directly type in an opening bracket, followed by an exclamation mark, the path to the object, and a closing bracket, for example `[!MyObject]`. To not show the current icon, but a certain icon of the object, type in the name or the number of the icon, separated by a comma.

```
[!Station1]
[!Station1, Default]
[!Station1, 0]
```

The result looks like this:



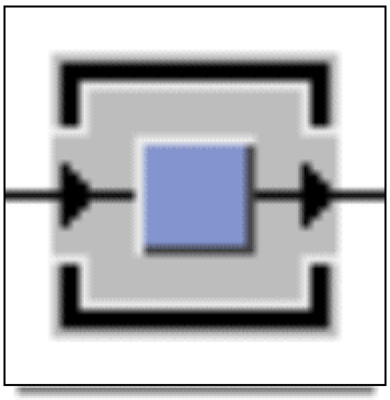
In addition, you can set the size of the icon as follows:

- Type in an asterisk * to also zoom the icon out, when you zoom the report window out. When zooming in, the icon is never enlarged to more than its original size.
- Type in a number and the percent sign % to scale the icon with this value.

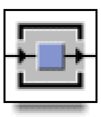
- Type in the width and the height of the icon in millimeters to show the icon with these measurements. You can also replace one of the two values with an *.

```
[!Station1, "Operational","My Station 50 x 50%", 50, 50]
[!Station1, "Operational", "My Station scaled to 110%", 110%]
```

The result looks like this:



My Station 50 x 50



My Station scaled to 110%

See also

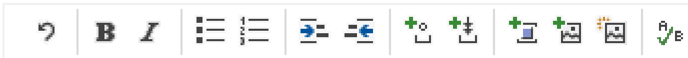
Add a Menu and Menu Commands

Show Statistics and Other Values in a Report

Add a Bitmap File

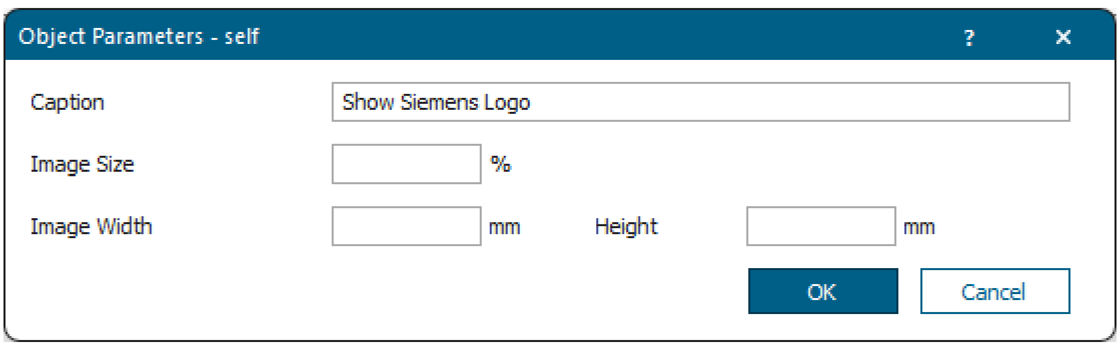
Specify the settings for creating an icon from a bitmap file and for displaying this icon on the HTML page.

Select the Object in the Dialog Select Object

To select a bitmap file to create an icon in this *HtmlReport* and to display this icon on the HTML page, click  on the toolbar . Select the bitmap file in the dialog **Open** and click **OK**.

All Files (*.bmp; *.dib; *.png; *.ppm; *.gif; *.dxf; *.dwg; *.ico)
Bitmaps (*.bmp; *.dib)
Portable Network Graphics (*.png)
Portable Pixelmaps (*.ppm)
GIF Files (*.gif)
Autodesk Drawings (*.dxf; *.dwg)
Icons (*.ico)
All Files (*.*)

Type in the object parameters:



In our example we selected the bitmap file named **SiemensLogo**. It is 31 mm wide and 6 mm high.

```
[!self, "SiemensLogo", "Show Siemens Logo"]
```

The result looks like this:´

See also

[Show Statistics and Other Values in a Report]

Collect Object Instances and Show Them

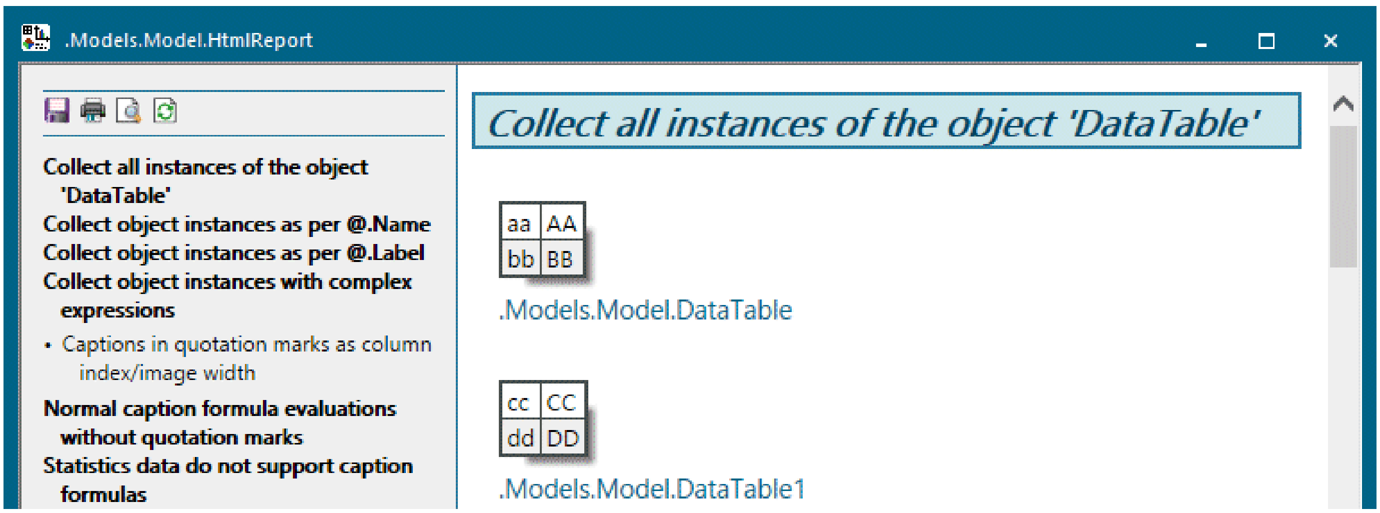
To insert all instances of a certain object into the *HtmlReport*, type in the path to the object followed by an asterisk * as parameter.

Remarks

Type in:

```
# Collect all instances of the object 'DataTable'
[.InformationFlow.DataTable*]
```

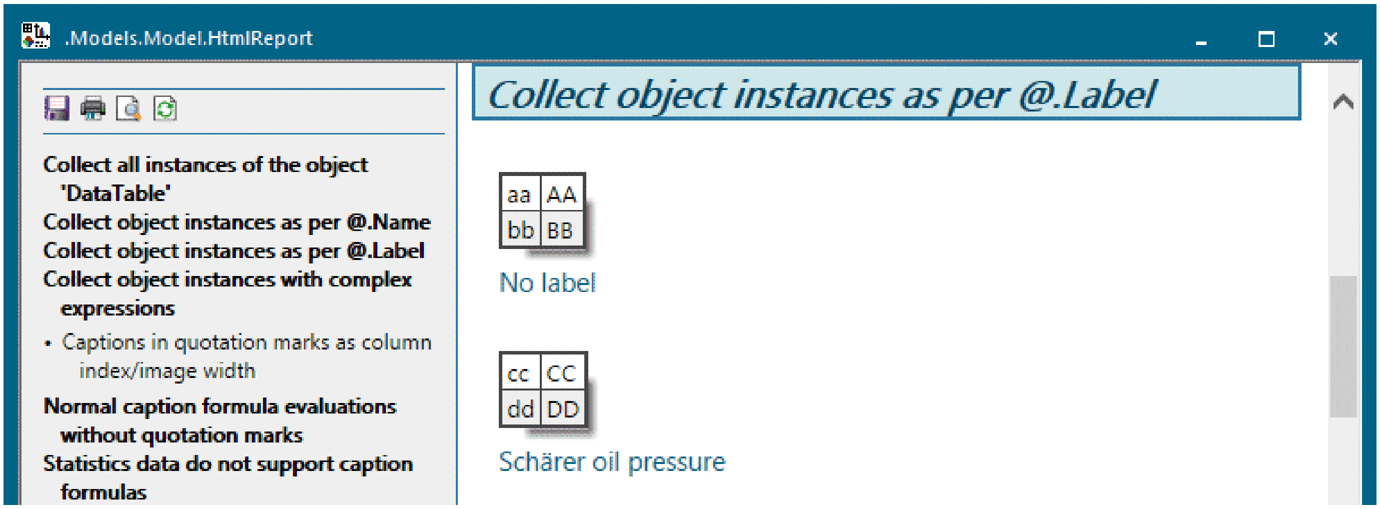
The result of this instruction looks like this:



The title in the *HtmlReport* can also contain a formula. In the simplest form it looks like this:

```
# Collect object instances as per @.Label
[.InformationFlow.DataTable*, "=@.Label"]
[.UserInterface.Chart*, "=@.Label"]
```

The result looks like this:



Start the formula, just like in the normal syntax of the *HtmlReport*, with `"="`. As the formula is enclosed in double quotation marks, you have to **protect** the double quotation marks within the formula itself with a backslash, for example `"=\"#\"`, to prevent it from being evaluated in the *HtmlReport* itself, so that the double quotation marks only matter in the formula itself.

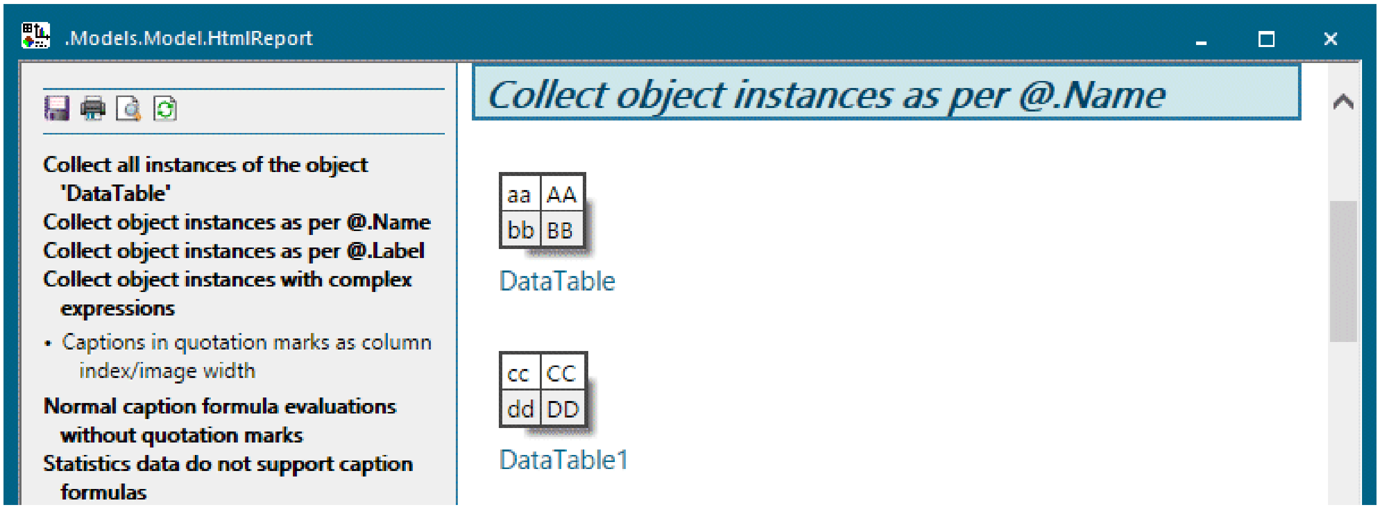
For the evaluation of the formulas you will use the usual anonymous identifiers where the anonymous identifier **@** always points to the instance to be shown.

If you access the name with `@.Name` or the label with `@.Label` in formulas and if either the name or the label is used for the title, *Plant Simulation* adds the name or the label as link to the *HtmlReport*. When you then click the link, *Plant Simulation* opens the instance.

Compare this example:

```
# Collect object instances as per @.Name
[.InformationFlow.DataTable*, "=@.Name"]
```

The result looks like this:



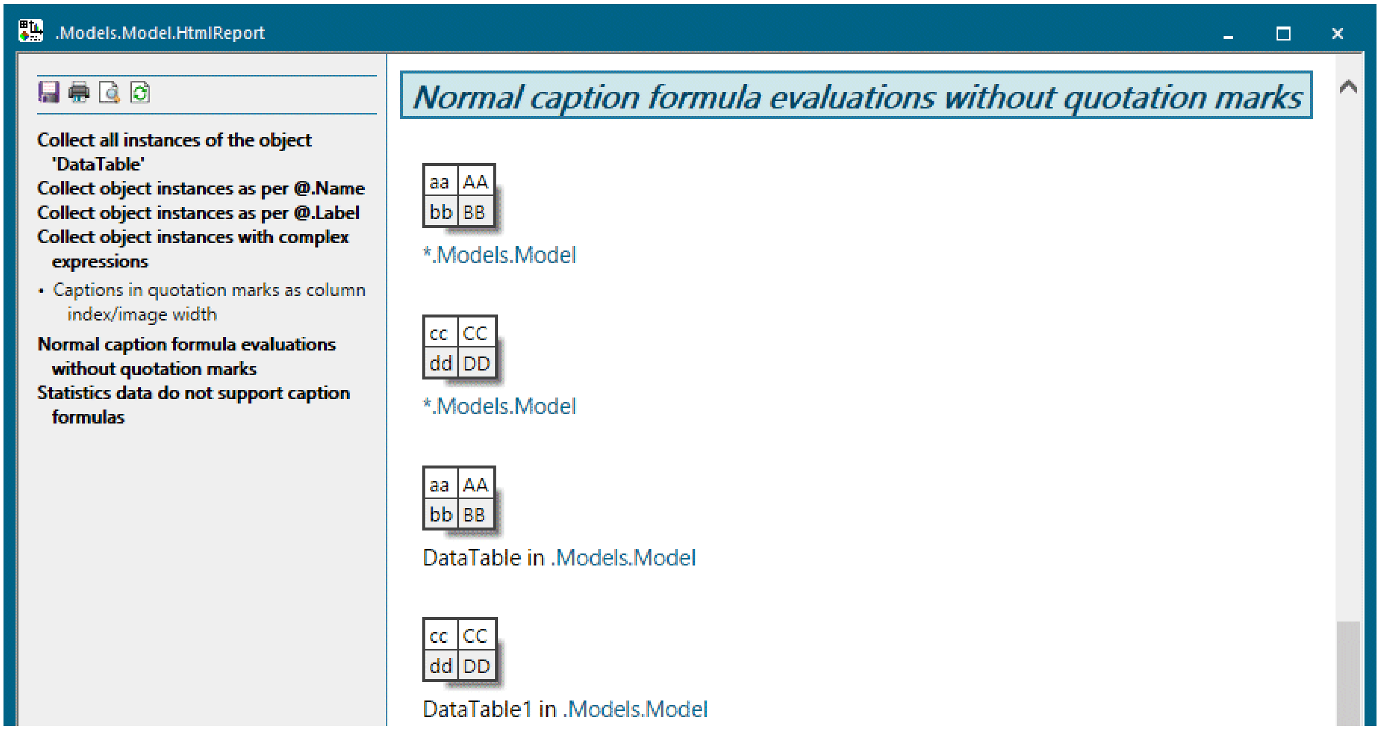
If the thus generated title contains a valid path to an existing object, *Plant Simulation* inserts the path as a link into the *HtmlReport*.

The following example generates an HTML table for each instantiated *DataTable* within the *Frame* in which the *HtmlReport* is located and a title with the absolute path to the instance of the *DataTable*. You can then click the underlined link to open this *DataTable* instance.

These instructions

```
# Normal caption formula evaluations without quotation marks
[.InformationFlow.DataTable*, "=@.Location"]
[.InformationFlow.DataTable*, "=@.Name + \" in \" + to_str(@.Location)"]
```

insert this into the *HtmlReport*:



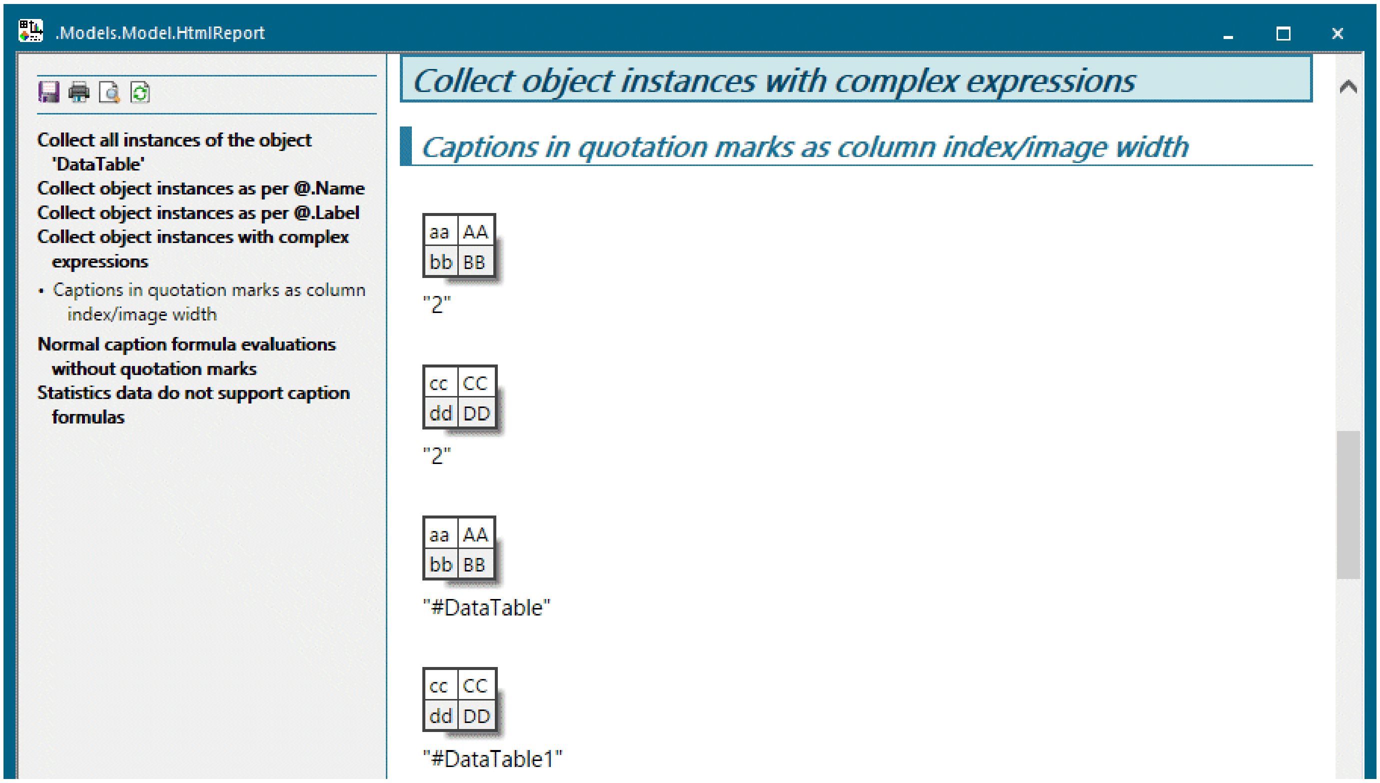
In general *Plant Simulation* underlines all titles below object depictions which contain a valid path. You can then click this link to open the respective object.

This also applies for titles which, among others, contain a valid path. Here *Plant Simulation* adds the path to the contained *Frame* for each *DataTable* in the title as a link.

These instructions

```
# Collect object instances with complex expressions
## Captions in quotation marks as column index/image width
[.InformationFlow.DataTable*, "=1+sin(pi/2)"]
[.InformationFlow.DataTable*, "=\"#\" + @.Name"]
[.UserInterface.Chart*,"=1+sin(pi/2)"]
```

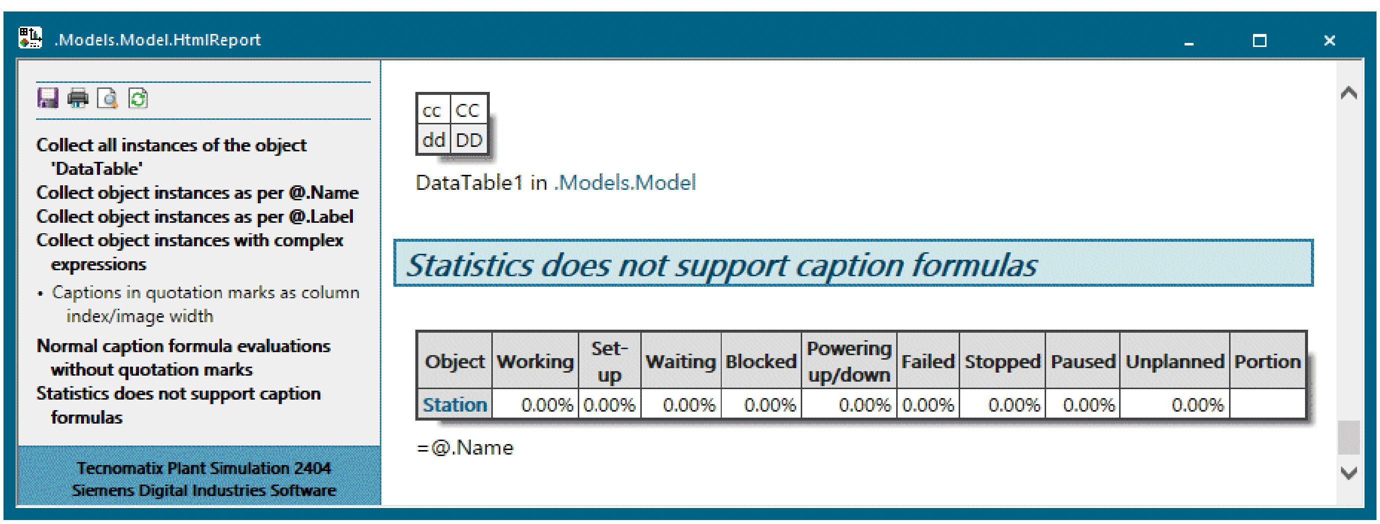
insert this into the *HtmlReport*.



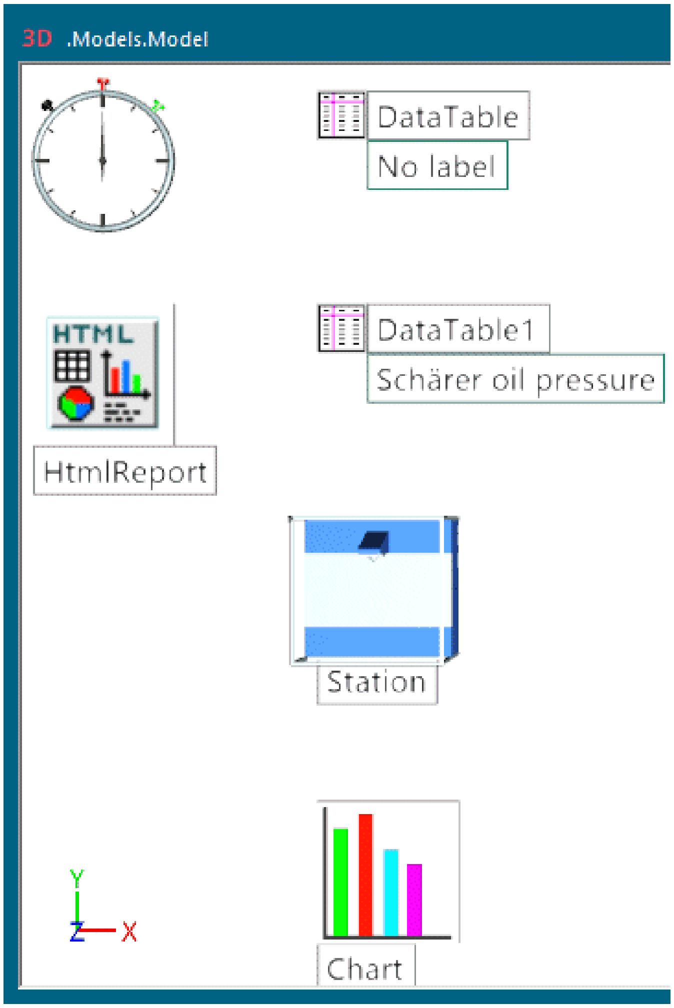
Statistics data do not support caption formulas. These instructions

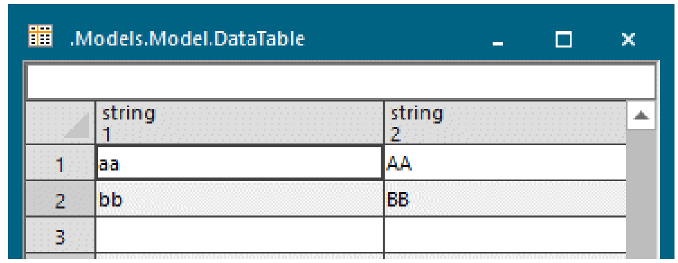
```
# Statistics data do not support caption formulas
[.MaterialFlow.Station*, "=@.Name"]
```

insert this into the *HtmlReport*:



This source code demonstrates the functions described above for our *Frame*. The *DataTables* have to contain data.






```
# Collect all instances of the object 'DataTable'
[.InformationFlow.DataTable*]

# Collect object instances as per @.Name
[.InformationFlow.DataTable*, "=@.Name"]

# Collect object instances as per @.Label
[.InformationFlow.DataTable*, "=@.Label"]
[.UserInterface.Chart*, "=@.Label"]

# Collect object instances with complex expressions
## Captions in quotation marks as column index/image width
[.InformationFlow.DataTable*, "=1+sin(pi/2)"]
[.InformationFlow.DataTable*, "=\"#\" + @.Name"]
[.UserInterface.Chart*,"=1+sin(pi/2)"]

# Normal caption formula evaluations without quotation marks
```

```
[.InformationFlow.DataTable*, "=@.Location"]
[.InformationFlow.DataTable*, "=@.Name + \" in \" + to_str(@.Location)"]

# Statistics data do not support caption formulas
[.MaterialFlow.Station*, "=@.Name"]
```

See also

Display a HtmlReport

Insert a Comment into Your Report

Type in two hyphens `--` to start a comment which extends to the end of the line.

Remarks

The comments will be ignored when generating the HTML page, meaning they do not affect the look of the HTML page. The comment is not part of the .htm file when you save the *HtmlReport*.

```
This text will be displayed. -- This text will not be displayed.
```

The result looks like this:

This text will be displayed.

See also

Show Statistics and Other Values in a Report

Insert a Horizontal Line

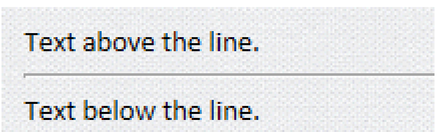
Type in three hyphens `---`, which are not followed by text, to create a solid horizontal line.

Remarks

You can also type more than three hyphens. If you type in other characters than the hyphens into the line of characters, *Plant Simulation* interprets `---` as the start of a comment.

```
Text above the line.
------
Text below the line.
--- This is a comment ---
```

The result looks like this:



See also

Show Statistics and Other Values in a Report

Insert HTML Tags

Type the HTML tags within angle brackets, according to HTML syntax.

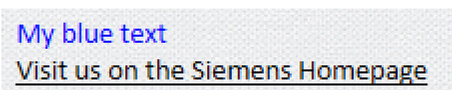
Remarks

Plant Simulation only recognizes the HTML tag if no space follows directly after the opening angle bracket (<), and only if the closing angle bracket (>) is located in the same line. If this is not the case, *Plant Simulation* displays a normal opening angle bracket.

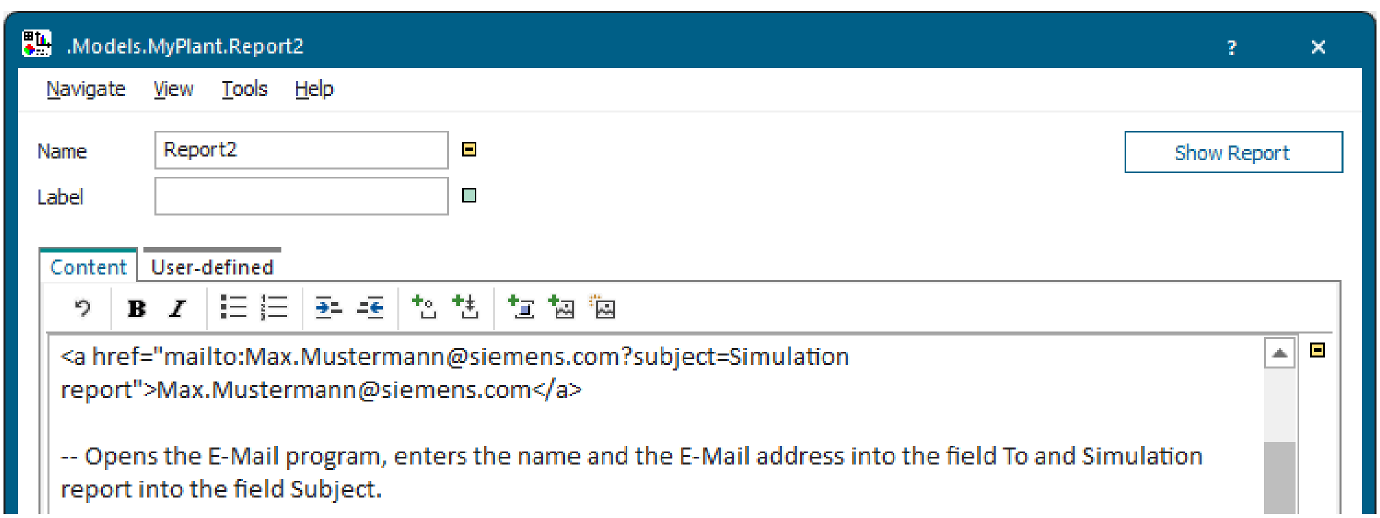
Plant Simulation does not analyze the markup language within the HTML tags. This way `*`, `1.`, `--`, etc. do not have any special meaning within the angle brackets.

```
<span style="color:blue">My blue text</span>
<a href="http://www.siemens.com">Visit us on the Siemens Homepage</a>
<!-- This is an HTML comment which is exported to the .htm file. -->
```

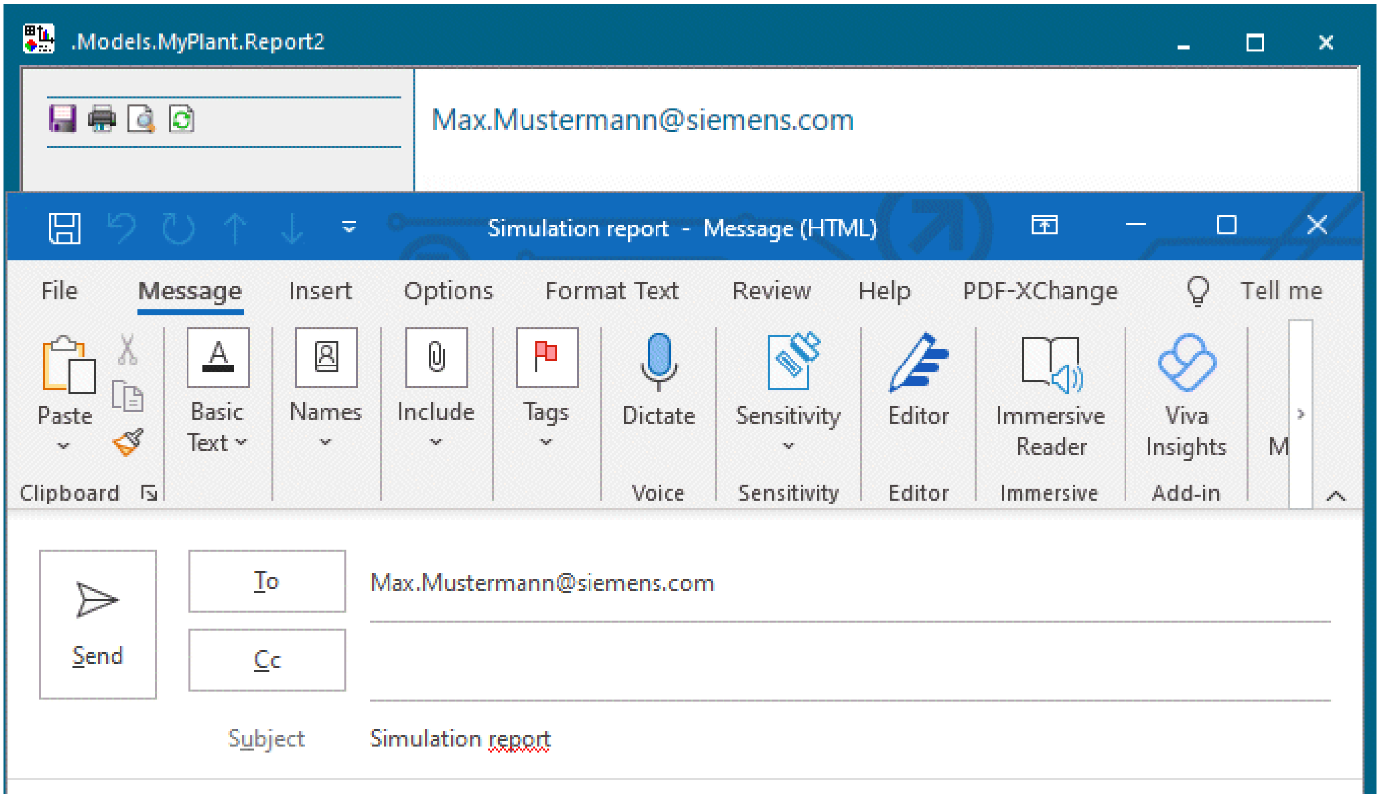
The result looks like this:




```
<a href="mailto:Max.Mustermann@siemens.com?subject=Simulation
report">Max.Mustermann@siemens.com</a>
-- Opens the E-Mail program, enters the name and the E-Mail address into
the field To and Simulation report into the field Subject.
```



The result looks like this:



See also

Show Statistics and Other Values in a Report

Protect Characters

When you type a character, which has a special meaning in the markup language, *Plant Simulation* does not display this character on the HTML page, but treats it as part of the notation of the markup language.

Remarks

To show such a character on the HTML page, type in a backslash \ to the left of the character.

```
\-- This is not a comment.
1\. This is not a numbered list.
\[%\] This is a percentage sign within brackets.
```

The result looks like this:

-- This is not a comment.
1. This is not a bulleted list.
[%] This is a percentage sign within square brackets.

You can protect the following special characters with a backslash \, but you do not have to protect them in each and every case:

Special character	Stands for	Type in to protect
\	backslash	\\
*	asterisk	*
#	number sign	\#
[	opening bracket	\[
]	closing bracket	\]
-	hyphen	\-
.	period	\.
<	less-than	\<
>	greater-than	\>

See also

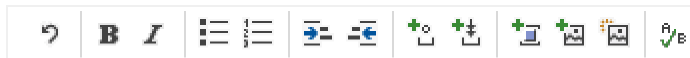
Show Statistics and Other Values in a Report

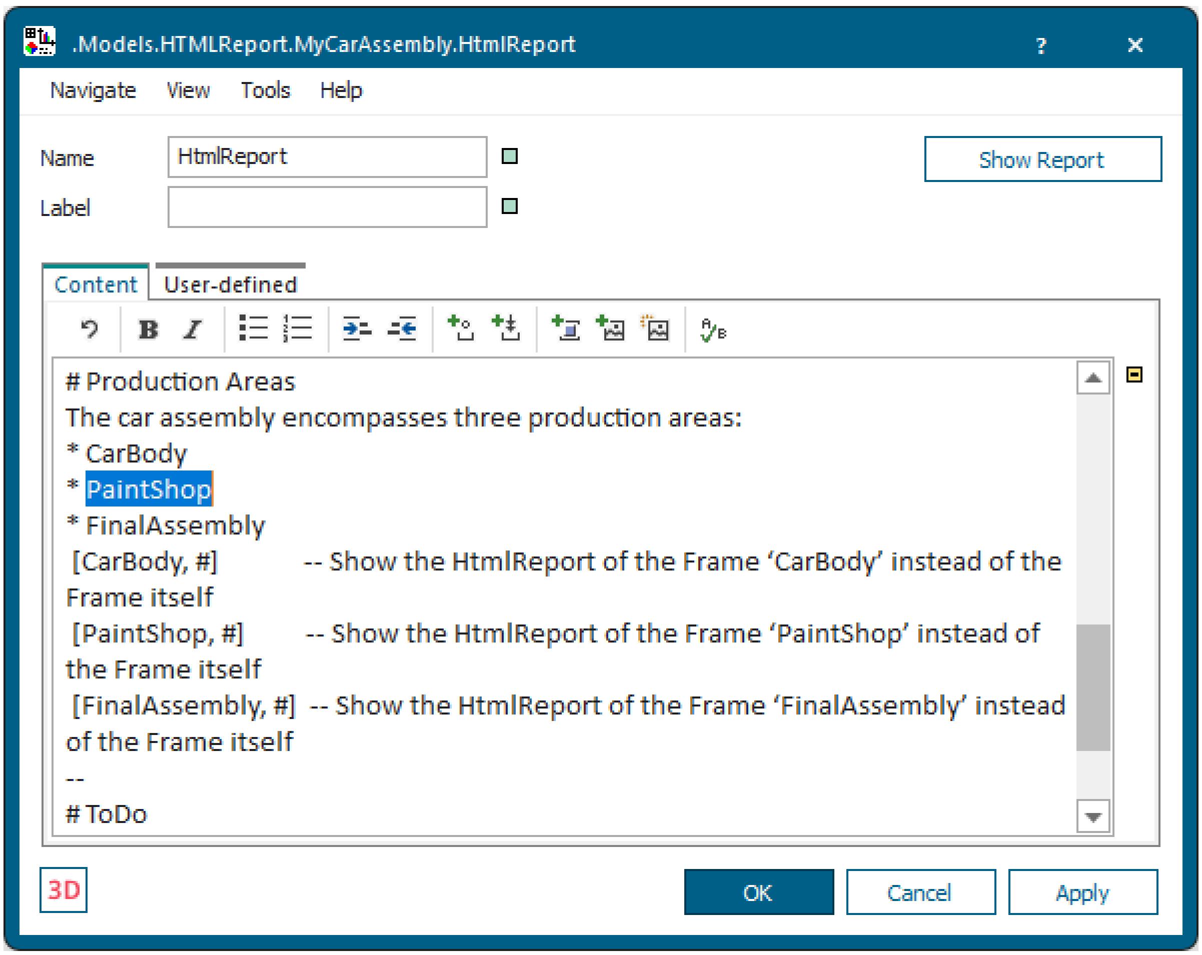
Spell Check the Content of the HtmlReport

To spell check the text on the tab **Content** starting at the cursor position, click the button. *Plant Simulation* then selects the next spelling error it found.

Remarks

Plant Simulation uses the Windows spell checker. It ignores text within '[' and ']'.

Instead of clicking  on the toolbar , you can also press **Ctrl+S**.



Note

The Windows spell checker does not work for Chinese and Japanese.

Use a String Unchanged in the HTML Code

To integrate a string into the HTML code without changing it, enclose it between two opening angle brackets << and two closing angle brackets >>.

Remarks

The string can have any length and can contain any number of line breaks.

This may be required for two reasons:

- To prevent that a long string of text is being interpreted as markup language.

  To prevent that a long string of text from being interpreted as markup language, you can type in a backslash `\`. This might prove to be cumbersome though. Imagine that you want to display the string `------` on the HTML page. To prevent two consecutive hyphens from being interpreted as comment, you would have to protect each second hyphen with a backslash: `-\--\--\-`. Instead, you can also enclose the string within two angle brackets: <<`------`>>.

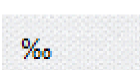
- To generate characters which have a special meaning in HTML.

  When you type in a character that has a special meaning in HTML, *Plant Simulation* automatically converts this character into an HTML entity. An example for a special meaning in HTML are the quotation marks (`"`). For this reason *Plant Simulation* generates the HTML entity **"** for quotation marks within the HTML code.

  If you want to generate a quotation mark within the HTML code (this way you can easily generate invalid HTML), you cannot accomplish this by protecting the quotation mark with a backslash. *Plant Simulation* will convert this quotation mark into an HTML entity to avoid generating invalid HTML code. To prevent this, you can enclose the quotation mark in two opening and two closing angle brackets <<“>>.

  Suppose that you want to display the per mille sign ‰ but that you cannot enter it by typing on the keyboard. If you know that the per mille sign has the Unicode code 8240, you can formulate the sign as the HTML entity `‰`. As the sign & has a special meaning in HTML (namely to open the HTML entities), *Plant Simulation* generates the HTML code **‰**, so that the Browser displays the string **‰**.

  If you want to display the per mille sign in the Browser, you have to enclose the string in two angle brackets so that *Plant Simulation* integrates it into the HTML code: <<`‰`>> without changing it.

See also

Show Statistics and Other Values in a Report

Tab User-defined

Define your own attributes as described under the **Tab User-defined**.

Navigate Menu

The commands are described under the **Navigate Menu**.

View Menu

The commands are described under the **View Menu**.

Tools Menu

The **Tools Menu** provides commands to access its functions.

Edit Controls

Edit Observers

See also

Tools Menu [general description]

Help Menu

The commands are described under the **Help Menu**.

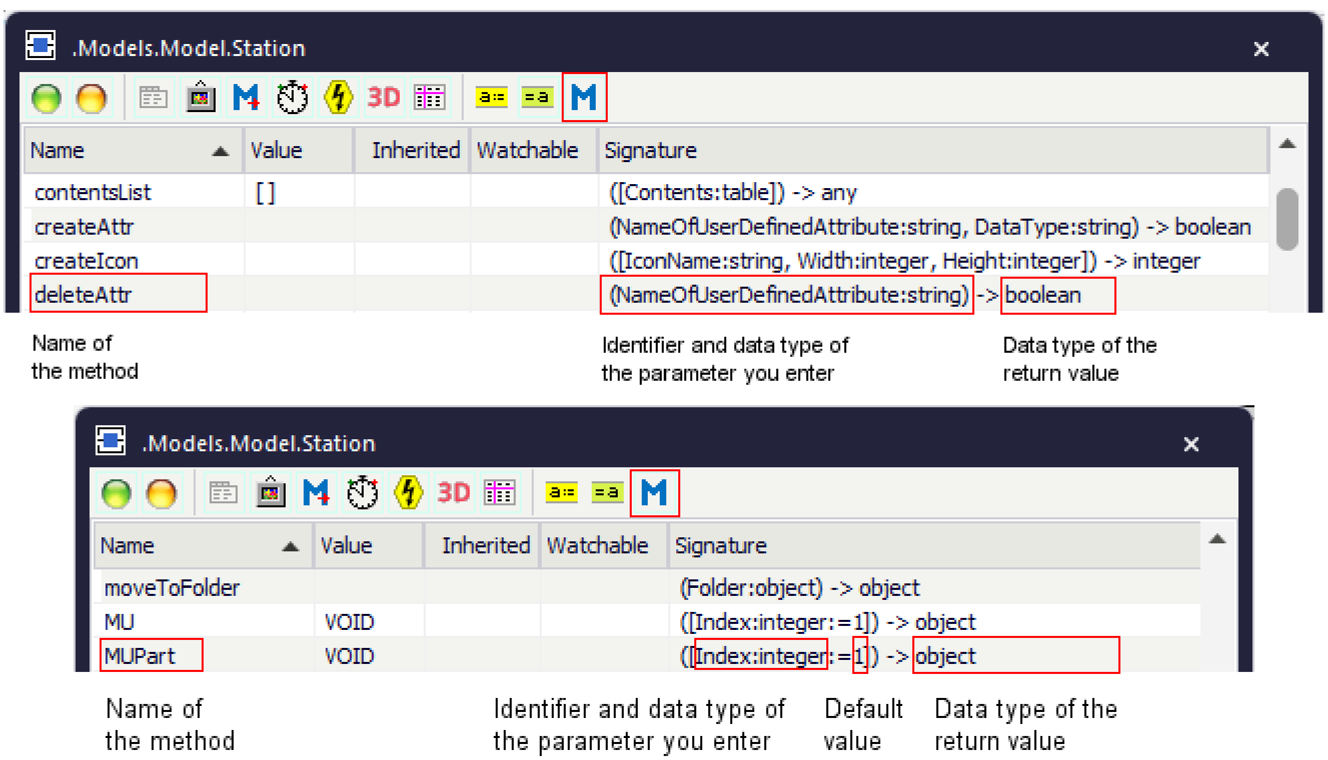
Methods of the HtmlReport

_Methods of the HtmlReport

The *HtmlReport* provides:

- The *methods* listed in the table of contents to the left.
- The **Methods of All Objects**.

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window **Show Attributes and Methods**. The figure below illustrates the information using the example of the object *Station*.



- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected **Class [general description]**.
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected **Instance [general description]**.